ABM ENGINE — BUSINESS LOGIC BIBLE

DOCUMENT 1 OF 3: MVP ENGINE

Functional Specification — Full Rule Compendium with Resolved Conflicts

Decimal Technologies | KSA Banking & Financial Institutions | June 2026

DOCUMENT 1 SCOPE — MVP ENGINE

This document contains every rule required to build a working ABM engine that proves one intelligent outbound loop. All 34 coherence-audit conflicts are resolved and their resolutions are integrated directly into the rules below. Each rule is presented in full Functional Specification style: the rule statement followed by its Rationale (the failure it prevents).

MVP delivers: Signal ingestion, account scoring, enrichment, AI reasoning + message generation, QC, human-reviewed email outreach, engagement tracking, sales handoff, CRM sync, and the core state machine.

MVP operating discipline: Autonomy capped at A1 (recommend and wait). Account-level scoring. Email-only channel. 10-20 target accounts. Every rule here ships in Phase 1 and requires no infrastructure beyond what Sections 1-4 specify.

DOCUMENT CONTENTS

Total rules in this document: 462

Section 1: Foundational Principles: Event-Driven + Enterprise Decision (132 rules)

ACCT Rules (15)

ACCT-COORD-001
When outreaching to multiple contacts at the same account, stagger initial touches by minimum 48 hours between contacts. Contact #1 Day 1, Contact #2 Day 3, Contact #3 Day 5.
Rationale: *Prevents the "everyone got emailed on the same day" problem that signals mass outreach rather than strategic engagement.*

ACCT-COORD-002
Maximum 3 contacts per account in active outreach simultaneously during MVP.
Rationale: *More than 3 creates coordination complexity and increases conflicting message risk.*

ACCT-COORD-004
Conflicting signals at the same account are flagged as "SIGNAL_CONFLICT." The system does NOT auto-resolve. Both signals are presented to the human reviewer. AI can suggest resolution but cannot execute autonomously.
Rationale: *Conflicting intelligence requires strategic judgment humans are better at.*

ACCT-COORD-005
Accounts can be linked as "related_accounts" with relationship type: PARENT_SUBSIDIARY, PARTNER, SHARED_BOARD, SAME_GROUP. When one account in a group reaches OUTREACH_ACTIVE, alert account_owner to manually coordinate.
Rationale: *Related accounts often have overlapping stakeholders — uncoordinated outreach across group entities looks uncoordinated.*

ACCT-COORD-006
— A contact appearing at TWO related accounts gets a "primary_account" designation. Outreach ONLY through their primary account.
Rationale: *Receiving outreach from two "different" Decimal campaigns simultaneously destroys credibility. Human-AI Review Logic*

ACCT-CORE-001
10-behavior Account-Centric Matrix: (1) Scoring at account level, (2) Tier at account level, (3) Outreach triggered by account evaluation then contact selection, (4) Reply from any contact pauses entire account, (5) Engagement rolls up to account, (6) One CRM opportunity per account, (7) Enrich account first then contacts, (8) AI reasons about organization + role, (9) Cooldown per account, (10) Analytics roll up to account.
Rationale: *Every integration defaults to contact-centric. These 10 behaviors must be explicitly enforced.*

ACCT-IDENTITY-001
Account = single legal entity or operationally independent business unit with its OWN CTO/IT leadership AND its OWN technology budget. If uncertain, default to SEPARATE accounts linked as related.
Rationale: *Merging loses granularity. Separating and linking preserves both independence and coordination.*

ACCT-IDENTITY-002
Account Lifecycle Events: MERGER: new merged account, migrate all data, archive old records, pause active sequences for review. ACQUISITION: acquired stays separate, relationship changes to PARENT_SUBSIDIARY, check if still independent procurement. SPIN-OFF: new account, copy relevant contacts, original keeps history. REBRAND: update name, add old name as alias.
Rationale: *Each lifecycle event requires different data handling.*

ACCT-IDENTITY-004
Geographic Scoping: MVP = KSA only. GCC subsidiary signals attributed to KSA parent at 70% relevance weight. UAE-HQ fintech opening KSA office = new account scoped to KSA, HQ signals weighted at 70%.
Rationale: *HQ decisions often flow to regional branches but aren't 100% applicable.*

ACCT-IDENTITY-006
Account Promotion: Watch->Active when P1/P2 signal + Fit >= 50. Active->Watch when COLD for 90+ days. Active->Archive only by manual human decision. Archive->Watch by manual decision.
Rationale: *Promotion is signal-driven. Demotion requires sustained inactivity. Archival requires human judgment. — 2F. Buying Committee Model (6 Rules)*

ACCT-COMMITTEE-004
Cross-Role Message Coherence: All messages to same account reference SAME signal, highlight SAME Decimal capability, propose SAME outcome. Only the ANGLE and PROOF POINTS differ by role. The NARRATIVE is consistent.
Rationale: *If CTO and Head of Lending compare notes, messages should tell same story from different angles.*

ACCT-COMMITTEE-006
Informal Influence Mapping: Each contact gets influence_modifier: STANDARD (matches title), ELEVATED (disproportionate influence — royal connection, CEO advisor, 15+ year tenure), REDUCED (less than title suggests — <6 months, interim, known to be leaving). Source: connector intel, BD knowledge, manual research only. ELEVATED contacts prioritized one seniority level higher.
Rationale: *Title does not equal power in KSA banking. — 2G. Failure Mode A: Account Misattribution (4 Rules)*

ACCT-PAUSE-001
Pause depends on reply type: POSITIVE (let's talk) = full account pause. REFERRAL (talk to [Name]) = pause this contact, redirect to referred person, continue others. NEGATIVE-PERSONAL (I'm not interested) = pause this contact, continue others unless reply says "our COMPANY isn't interested." NEGATIVE-ORGANIZATIONAL (we chose a vendor/no budget) = full pause + human review. AUTO-REPLY = no pause, resume after return. NEUTRAL (let me check) = pause this contact temporarily, slow others 50%.
Rationale: *Blanket pause for every auto-reply freezes the pipeline unnecessarily.*

ACCT-PAUSE-002
When system decides partial pause (not full): decision logged, account_owner notified, 4-hour override window. No override within 4h = decision stands.
Rationale: *Nuanced pause decisions need human oversight without requiring human initiation.*

ACCT-PAUSE-003
Pause Duration Limits: Day 7: reminder to account_owner. Day 14: escalate to manager. Day 21: forced decision — (a) "still in conversation" + reset timer, (b) "stalled" + revert to HOT + re-enable others, (c) "dead" + DORMANT.
Rationale: *Prevents "engaged zombie" accounts sitting paused forever because a human dropped the ball.*

AIBOUNDARY Rules (3)

AIBOUNDARY-001
"AI Permission Matrix" — Exhaustive CAN/CANNOT list: CAN: gather public info, calculate scores, generate drafts, tailor messaging, recommend strategy, classify signals/sentiment/outcomes, analyze performance, validate outputs, propose send times, prepare CRM data. CANNOT: send messages without approval (MVP), change scoring formulas/weights/thresholds, reclassify human-verified items, override its own QC rejections, override business hours/holidays, create/modify/delete CRM deals, access private systems, execute strategy changes without confirmation. DEFAULT for unlisted actions: AI CANNOT.
Rationale: *Without explicit boundaries, AI scope creeps.*

AIBOUNDARY-002
"Approved Claims Library" — AI can ONLY make claims about Decimal from a pre-approved library: verified client references (REFERENCEABLE only), specific product capabilities (reviewed by product team), performance metrics (verified by data), compliance certifications (verified by legal), integration capabilities (verified by engineering). Claims not in library = "CLAIM_NOT_VERIFIED" flag + placeholder. AI must NEVER: guarantee outcomes ("WILL increase lending 40%"), make comparative claims ("better than Temenos"), claim non-existent capabilities, reference NDA-protected info.
Rationale: *One false capability claim to a bank CTO creates compliance and trust crisis.*

AIBOUNDARY-003
"Cultural Sensitivity Rules" — KSA-specific norms hardcoded: FORMALITY: formal titles always in initial outreach, never initiate informality. RELIGION: never reference practices/beliefs unless contextually appropriate (Ramadan greetings OK during Ramadan). GENDER: no assumptions about capabilities based on gender, same professional tone regardless. HIERARCHY: never suggest lower-ranking should "convince" superiors — frame as "sharing information." NATIONAL PRIDE: frame Decimal as supporting Vision 2030, never as "teaching Saudi banks." COMPETITOR RESPECT: respectful tone even when positioning against competitors.
Rationale: *Cultural missteps in KSA banking are relationship-ending. DEEP DRILL: Failure Modes 1-6 (11 Rules) FM1 — Rubber-Stamping (2 Rules)*

AIHUMAN Rules (5)

AIHUMAN-FOUNDATION-001
"The AI-Human Boundary Principle": AI operates in the domain of DATA (signals, scores, research, patterns, drafts). Humans operate in the domain of RELATIONSHIPS (trust, judgment, negotiation, reputation). The boundary between these domains shifts as AI proves reliability, but the RELATIONSHIP domain is permanently human. The system must be designed so AI never takes actions in the relationship domain without explicit human authorization.
Rationale: *Enterprise banking sales in KSA are trust-based, relationship-based, and culturally nuanced. A single AI hallucination to a SEVP at SNB could permanently close the door. The cost of one bad message outweighs 100 automated good ones.*

AIHUMAN-FOUNDATION-002
"The 90/10 Resource Model": AI handles approximately 90% of work volume (signal monitoring, scoring, enrichment, research, reasoning, drafting, QC, engagement tracking, analytics). Humans handle approximately 10% (message approval, relationship conversations, strategic overrides, handoff management, outcome classification). But the 10% human contribution carries 90% of the REPUTATIONAL RISK. Therefore: every human decision point must be presented with complete AI-prepared context so the human can make a high-quality judgment in minimum time.
Rationale: *Without AI, a BD person spends 2-3 hours researching per account. With AI, they spend 5 minutes reviewing a strategically reasoned draft.*

AIHUMAN-CONTEXT-001
"Human Decision Context Package" — Every human decision point must present complete context: (1) Message Approval: full reasoning chain, triggering signal with source, account summary, persona profile, chosen strategy with rationale, Golden Window status, conflict flags, ICS score, Effective Opportunity. (2) Sequence Strategy: contacts in order, channel reasoning, timing rationale, competitive landscape, relationship paths, committee coverage, blocker/veto flags. (3) Handoff Response: full engagement history, reply text with sentiment, meeting brief, talking points, likely objections, comparable wins. (4) Override Decision: system recommendation with rationale, what changes if overridden, historical override outcomes, risk assessment. (5) Outcome Classification: sequence timeline, engagement events, contact behavior, signal quality assessment, AI suggested classification.
Rationale: *"Approve t...*

AIHUMAN-DISAGREE-001
"Disagreement Resolution Protocol": The human ALWAYS wins during MVP. But disagreement must be CAPTURED: (a) record AI output, human change, human stated reason (required, min 10 chars), (b) data feeds three mechanisms — prompt improvement (humans change ANGLE → persona prompts need updating), knowledge gap (humans add facts AI didn't know → enrichment incomplete), style mismatch (humans change tone/length → generation constraints need adjusting), (c) monthly report: "Top 10 reasons humans overrode AI" — single most valuable input for AI improvement, (d) CRITICAL: system must NEVER train AI to simply mimic human edits without understanding WHY. A human might edit for situation-specific reasons that shouldn't generalize.

AIHUMAN-DISAGREE-002
"AI Guardrails on Human Actions" — AI doesn't override humans but CAN warn: SOFT WARNINGS (overridable, logged): message contains unverifiable claims ("Source not found in enrichment"), edit introduces banned phrase or exceeds length, human force-promotes account with ICS <40. HARD BLOCKS (require escalation): message would send outside KSA business hours or during Ramadan/Eid, message targets DISQUALIFIED or EXHAUSTED contact, message to DORMANT account in cooling-off period.
Rationale: *Humans make mistakes too — especially tired reviewers at 5pm Thursday. DEEP DRILL: Explainability Layer (5 Rules)*

CHANNEL Rules (1)

CHANNEL-GAP-002
CONFLICT RESOLUTION (FIN-C03): Minimum gap between two different-channel touches to the SAME contact is 24 hours, unless the contact engaged on the first channel (open, click, accept), in which case the second channel may follow after 4 hours. This is DISTINCT from ACCT-COORD-001, which governs spacing across different contacts at one account (48h stagger). Rationale: closes the coverage seam between across-contact and same-contact channel coordination.

ETHICS Rules (3)

ETHICS-001
"Ethical Absolutes — What System Must Never Do": (a) NEVER exploit personal vulnerability — mark contact "PERSONAL_SENSITIVITY" if intelligence reveals personal difficulty. (b) NEVER misrepresent identity — messages sent on behalf of named Decimal employees, no fake personas. (c) NEVER manufacture urgency — reference REAL urgency only, no "offer expires Friday" when it doesn't. (d) NEVER disparage competitors by name in automated outreach. (e) NEVER use engagement data punitively — don't escalate to boss or cc colleagues. (f) NEVER ignore opt-out — ABSOLUTE and PERMANENT unless explicitly reversed by contact.
Rationale: *The system manipulates human attention. That power requires ethical constraints.*

ETHICS-002
"Data Ethics": (a) Proportional collection: only data serving defined business purpose. (b) Data minimization in prompts: agents receive only needed data. (c) Retention limits: ARCHIVE accounts 12+ months → purge/anonymize personal contact details. Organizational intelligence retained. (d) Transparency on request: if contact asks "how did you get my info?" answer honestly. (e) PDPL compliance: quarterly review against Saudi data protection requirements.
Rationale: *Legal compliance and moral responsibility for personal data handling.*

ETHICS-003
"Personalization vs Manipulation Line": ACCEPTABLE: "Given your focus on digital lending growth..." (real business priority). "Banks that modernized saw 40% improvement..." (factual, relevant). UNACCEPTABLE: "Your competitors are already ahead" (manufactured fear, may not be true). "You can't afford to wait" (pressure not grounded in deadline). "As a CTO who cares about their legacy..." (identity manipulation). "Other executives at your bank are interested" (internal social proof, may be false and politically dangerous). THE TEST: "Would recipient feel RESPECTED seeing the AI reasoning?" If they'd feel analyzed/profiled/manipulated → crosses the line. Agent 4 produces BUSINESS pains, never insecurities. Agent 5 selects VALUE angles, never fear. Agent 6 writes with RESPECT, never urgency manufacturing.

EVENT Rules (4)

EVENT-RESP-001
"Immediate" for P1 signals means scoring recalculation must BEGIN within 15 minutes of signal ingestion. For P2, within 1 hour. The entire cascade (scoring, tier check, enrichment trigger) must complete within 2 hours for P1 and 6 hours for P2.
Rationale: *P1 signals have a golden window. A new CTO spends weeks 1-4 evaluating the landscape. 15 minutes is fast enough to stay same-day. 1 hour for P2 is sufficient for strategic-but-not-urgent signals.*

EVENT-RESP-002
Signal ingestion and scoring run 24/7/365 regardless of business hours. Only OUTREACH EXECUTION respects business hours and KSA weekends.
Rationale: *A P1 signal at 11pm Thursday should have score updated, enrichment triggered, AI reasoning complete, and message drafted by Friday morning, ready to send Sunday 9am.*

EVENT-RESP-003
The system maintains a "Ready-to-Fire Queue" for messages that have passed AI QC and await either human review or next valid send window. Queue ordered by signal priority (P1 first) then signal age (oldest first within same priority).
Rationale: *Ensures highest-priority messages send first when business hours resume.*

EVENT-RESP-004
Every event carries a "trace_id" linking the original trigger through every downstream action: signal, scoring record, enrichment record, AI generation, message, and engagement.
Rationale: *If anyone asks "why did this person receive this message?" the trace_id answers in one lookup. Full audit trail. Account Coordination*

EXPLAIN Rules (5)

EXPLAIN-001
"Reasoning Chain Transparency" — Every AI-generated message carries a "Reasoning Passport": SIGNAL: text, date, source, confidence. INTERPRETATION: what it means for buying intent (Agent 1). ACCOUNT CONTEXT: organization summary (Agent 2). PERSONA ANALYSIS: role motivations, decision criteria (Agent 3). PAIN INFERENCE: likely operational pains with reasoning (Agent 4). STRATEGY: chosen angle, value proposition, CTA rationale (Agent 5). QC RESULT: all checks passed, confidence score (Agent 7). Stored permanently alongside message. Visible to reviewer. Part of handoff package.
Rationale: *If you can't explain why you sent a message, you shouldn't send it.*

EXPLAIN-002
"Score Decomposition" — Every score breakable into contributing factors: Dynamic Score 82 = Signal Strength 75 (weight 35%, contribution 26.25) + Regulatory Pressure 90 (30%, 27) + Reachability 70 (20%, 14) + Warmth 60 (15%, 9) + intent multiplier + criticality bonus. ICS decomposed similarly. Effective Opportunity shows each multiplier. Every number has traceable source. No black boxes.
Rationale: *When human asks "why is this account #3 not #1?" system shows exactly which dimensions differ.*

EXPLAIN-003
"Negative Decision Logging" — Every time system decides NOT to act, it logs: "Account X not promoted to HOT because ICS = 35 (below 40 per GOV-001)." "Message for Contact Y blocked: OUTREACH_EXHAUSTED (per FAIL-FATIGUE-002)." "Outreach deferred: Golden Window expired, needs regeneration (per TIMING-WIN-002)." These logs are queryable. Owner asks "Why hasn't anything happened with Bank X?" and gets a specific answer.
Rationale: *Silent inaction is the hardest thing to debug. Negative decisions must be as visible as positive ones.*

EXPLAIN-004
"Human Decision Audit Trail" — Every human action logged with: WHO (which human), WHAT (approved/edited/rejected/overrode/escalated), WHEN (timestamp),
Rationale: *(stated reason — required for edits/rejections/overrides), AGAINST (what AI recommended if different), OUTCOME (what happened as result). Creates complete audit trail where BOTH AI reasoning and human judgment are visible. WHY: If a message causes a problem, the trail shows: AI recommended X, human changed to Y, for reason Z. Accountability is clear.*

EXPLAIN-005
"Accountability Framework" — 4 layers: Layer 1 — AI QC Agent: accountable for catching hallucinations, tone violations, factual errors. Layer 2 — Human Reviewer: accountable for strategic judgment (right angle? right timing? right tone for relationship?). Layer 3 — Account Owner: accountable for relationship outcomes. Layer 4 — System Designer (BD leadership): accountable for rules, thresholds, workflows, guardrails. Post-mortem identifies WHICH layer failed. No finger-pointing between AI and humans.
Rationale: *Clear accountability prevents blame-shifting and drives targeted fixes. DEEP DRILL: AI Behavioral Boundaries (3 Rules)*

FAIL Rules (67)

FAIL-SILENT-001
"Signal Health Score" per source. Track: last_successful_poll, signals_returned_count, error_count, consecutive_zero_result_polls. If any source returns zero results for 3 consecutive poll cycles, trigger alert: "SOURCE_POSSIBLY_DOWN."
Rationale: *The system cannot distinguish between "no news today" and "our connection is broken" without monitoring.*

FAIL-SILENT-003
Every poll cycle must log: source, start_time, end_time, http_status, signals_found, errors. Even "zero results" is a valid log. Missing log = poll didn't run (different failure).
Rationale: *Distinguishes "polled and found nothing" from "poll never executed."*

FAIL-SILENT-004
"Stagnation Detection" — If a HIGH-FIT account (Static Fit Score >= 60) has been WATCHING with zero signals for 45 days, flag as "SIGNAL_DARK_ACCOUNT" for weekly human review. Human can: (a) manually research, (b) broaden keywords, (c) accept silence, (d) force-promote.
Rationale: *Good accounts can go quiet. The system must surface them rather than letting them rot.*

FAIL-SILENT-005
Every account has TWO scores: (1) DYNAMIC SCORE (0-100) — signal-driven, changes with events, drives tier assignment. (2) STATIC FIT SCORE (0-100) — ICP match regardless of signals. Calculated from: segment fit, size fit, geography fit, technology gap, regulatory exposure. Set at account creation, updated during enrichment.
Rationale: *Intent (dynamic) and fit (static) are orthogonal. A perfect-fit account with no signals is still worth monitoring.*

FAIL-SILENT-006
Static Fit Score governs: (a) account entry threshold (only Fit >= 40 enters system), (b) tie-breaking priority, (c) stagnation detection threshold (only >= 60), (d) enrichment depth (Fit >= 80 gets FULL enrichment even at WARM).
Rationale: *Not all accounts deserve equal system resources. Fit determines baseline investment.*

FAIL-FALSE-001
Two-stage signal classification: Stage 1 — Keyword Detection (automated, catches potential signals). Stage 2 — Contextual Validation (AI reads full content and answers: "Does this indicate technology evaluation in categories Decimal serves?") Output: YES (proceed), MAYBE (proceed, confidence -0.2), NO (discard with log). Both stages must pass.
Rationale: *Keywords match context-free. AI validates actual meaning.*

FAIL-FALSE-002
Confidence score thresholds: 0.8-1.0 HIGH: full processing. 0.6-0.79 MODERATE: full processing, flag VERIFY_IF_POSSIBLE. 0.4-0.59 LOW: scoring only (reduced weight), NOT primary outreach trigger. 0.0-0.39 INSUFFICIENT: store for history, exclude from scoring.
Rationale: *Different confidence levels require different treatment.*

FAIL-FALSE-003
"Signal Retraction" workflow: A signal can be marked FALSE_POSITIVE by a human at any time. When retracted: (1) mark is_false_positive, (2) recalculate score without it, (3) if tier drops, downgrade, (4) if outreach was active, flag for human review, (5) if message already sent referencing this signal, prepare recovery brief for account_owner, (6) feed into AI classifier training.
Rationale: *False signals that trigger outreach can cause reputation damage. Recovery must be structured.*

FAIL-FALSE-005
"False Signal Pattern Library" — After 90 days, analyze confirmed false positives and identify patterns. Initial seed: app redesigns vs platform-level change, marketing partnerships vs tech deals, branch staff hiring vs tech hiring, sanctions/AML vs banking tech mandates, competitor press releases disguised as industry news. Patterns become negative rules in AI classifier (confidence -0.3).
Rationale: *Systematic pattern recognition prevents repeated mistakes. — 1D. Failure Mode 3: Signal Detected Too Late (6 Rules) The signal was real and high-intent, but the system found it too late. The Golden Window closed. Competitors got there first.*

FAIL-LATE-001
"Time-to-Detection" metric: time between event's ACTUAL date and system's detected_at timestamp. Target: median 48h for P1 in any 30-day period, trigger polling frequency review and source expansion evaluation.
Rationale: *Detection delay directly reduces outreach effectiveness.*

FAIL-LATE-002
"Manual Signal Fast Lane" — Human-entered manual signals are PRE-VALIDATED at confidence 0.9, skip AI classification Stage 2, enter scoring immediately. 24-hour processing SLA.
Rationale: *The human IS the classifier. Manual intelligence from BD team and connectors is high-reliability and time-sensitive.*

FAIL-LATE-004
"End-to-End Pipeline SLA" — Total elapsed time from signal detection to message sent: P1 target <= 24h to review queue, <= 48h to sent. P2: <= 48h to queue, <= 72h to sent. P3: <= 5 business days. P4: no SLA. Weekly bottleneck report.
Rationale: *Each stage adds latency. Without end-to-end SLA, no one owns the total delay.*

FAIL-LATE-005
"Emergency Enrichment" for P1 signals on accounts with no existing contacts: Abbreviated 3-step process: (1) find top 2 contacts by seniority via Apollo, (2) verify emails, (3) generate minimal account summary. Target: under 2 hours. Full enrichment runs in background.
Rationale: *P1 signals on unenriched accounts would otherwise lose their Golden Window waiting for full 7-step enrichment.*

FAIL-LATE-006
"Stale Signal Outreach Decision" — If signal detected AFTER Golden Window closed: P1: still process, but AI must NOT reference as "recent" — frame as ongoing initiative. P2: process only if another FRESH signal exists. P3/P4: context only, no outreach trigger.
Rationale: *Late detection doesn't invalidate the signal, but messaging must be reframed to avoid looking out of touch. — 1E. Failure Mode 4: Signal Without Buying Intent (5 Rules) The signal is real and timely, but doesn't actually indicate technology procurement. 'Record profits' doesn't mean they're buying a lending platform.*

FAIL-INTENT-001
Signals classified on TWO dimensions: RELEVANCE (P1-P4, already exists) and INTENT: ACTIVE_PROCUREMENT (direct vendor evaluation evidence), BUILDING (internal tech build evidence), STRATEGIC_SHIFT (change preceding procurement), CONTEXTUAL (relevant info, no procurement intent). Only ACTIVE_PROCUREMENT and BUILDING can be P1. CONTEXTUAL is P3/P4 max.
Rationale: *Priority without intent creates false urgency.*

FAIL-INTENT-002
Intent-based scoring multiplier: ACTIVE_PROCUREMENT x1.5, BUILDING x1.2, STRATEGIC_SHIFT x1.0, CONTEXTUAL x0.5. A P2 ACTIVE_PROCUREMENT signal (25 x 1.5 = 37.5 pts) scores HIGHER than a P1 CONTEXTUAL (40 x 0.5 = 20 pts).
Rationale: *Intent quality matters more than signal category.*

FAIL-INTENT-003
"Account-Relative Intent" — Same signal gets different intent based on account readiness. Open banking mandate: account readiness 1-3 = ACTIVE_PROCUREMENT (must buy). Readiness 4-6 = BUILDING. Readiness 7-10 = CONTEXTUAL (already compliant). If readiness unknown, default to STRATEGIC_SHIFT.
Rationale: *Same regulation creates different urgency depending on the bank's current capabilities.*

FAIL-INTENT-005
"Signal Accumulation Rule" — 3+ CONTEXTUAL signals for same account within 30 days = auto-upgrade ONE to STRATEGIC_SHIFT.
Rationale: *Multiple contextual signals (profits + hires + conference + press) in short window suggest expansionary phase. Individually none implies buying. Together they suggest growth investment. Cap: once per 30-day window. — 1F. Failure Mode 5: Signal Overload (6 Rules) Too many signals. 20 accounts score HOT. Daily cap is 5-7. Human reviewer has 50 messages. Everything is urgent. Nothing gets attention.*

FAIL-OVERLOAD-001
"Surge Detection" — If P1+P2 signals in any 24h window exceed 3x the trailing 7-day daily average, declare "SIGNAL_SURGE": (a) increase daily cap from 5 to 10, (b) assign second reviewer, (c) log surge event, (d) revert when volume <1.5x average for 3 days.
Rationale: *SAMA regulatory announcements can trigger 15+ accounts simultaneously.*

FAIL-OVERLOAD-002
"Surge Prioritization Stack Rank" — When more HOT accounts than capacity: Rank 1: ACTIVE_PROCUREMENT intent. Rank 2: relationship_warmth >= WARM. Rank 3: highest Static Fit. Rank 4: highest Dynamic Score. Rank 5: earliest Golden Window expiry. Serve top of stack until capacity reached.
Rationale: *Not all HOT accounts are equally likely to convert.*

FAIL-OVERLOAD-003
"Reviewer Load Balancing" — Max 20 messages/reviewer/day. Beyond this, quality degrades. Queue exceeds (reviewers x 20) = "REVIEWER_OVERLOAD" alert. Options: activate backup, extend P3/P4 SLA from 4h to 8h, auto-approve P3/P4 if Phase 2+ automation active. P1/P2 and C-suite: NEVER skip human review.
Rationale: *Reviewer fatigue produces approval rubber-stamping.*

FAIL-OVERLOAD-004
"Surge Differentiation Rule" — When 5+ accounts triggered by SAME signal: AI receives explicit instruction to differentiate. QC Agent checks: "Does this message contain at least 2 sentences specific to THIS account?" Fail = regenerate with stronger account-specific context.
Rationale: *15 banks getting the same SAMA-triggered message with only the name changed looks like mass marketing.*

FAIL-OVERLOAD-005
"Capacity State Machine" — System operates in: NORMAL (standard caps), SURGE (elevated caps, triggered by FAIL-OVERLOAD-001), DROUGHT (expanded signal search, triggered when signals NORMAL requires 3 days sub-threshold.
Rationale: *Different operating modes require different system configurations.*

FAIL-OVERLOAD-006
"Surge Retrospective" — Within 7 days of surge ending: How many accounts served vs queued? Average queue wait time? Golden Windows missed? Reviewer turnaround? Were surge-period messages lower quality (lower reply rate)?
Rationale: *Post-surge analysis feeds capacity planning improvements. — 1G. Failure Mode 6: Signal Manipulation (4 Rules) Companies publish fake signals. A CEO's conference speech about 'digital transformation' may be PR with no budget, no project, no real initiative. Public relations is not buying intent.*

FAIL-MANIP-001
"Observability Score" — Every signal gets evidence-based corroboration assessment: Press release ONLY = 30. Press + hiring = 60. Press + hiring + leadership hire = 80. Press + hiring + leader + RFP = 95. Job postings WITHOUT press release = 70 (stealth build, often more genuine). SAMA mandate + bank below compliance = 90. Conference presentation only = 25.
Rationale: *Single-source PR is unreliable. Corroborating evidence from independent sources validates intent.*

FAIL-MANIP-002
If average Observability Score across active signals <40, ICS Verification dimension is capped at 30 regardless of email verification quality.
Rationale: *Verified email to the right person is worthless if the REASON for reaching out (the signal) is unverifiable PR.*

FAIL-MANIP-003
"Signal Maturation Tracking" — Low-observability signals (<=40) enter "WATCH_FOR_CORROBORATION" state. If corroborating signal from different source arrives within 60 days: upgrade observability to average of both, recalculate ICS, log maturation.
Rationale: *The system doesn't reject PR signals. It withholds judgment and waits for evidence.*

FAIL-MANIP-004
"Account Credibility Index" (after 12 months of data) — Track whether signals at each account lead to actual activity: (Signals leading to verified activity within 6mo) / (Total signals). Credibility >= 70%: +0.1 confidence bonus. 40-69%: neutral. <40%: -0.15 confidence penalty — "this bank talks more than it acts."
Rationale: *Some organizations are chronic PR-without-substance offenders. — 1H. Failure Mode 7: Dark Intent — Buying Without Signals (5 Rules) The most dangerous blind spot. A bank internally approved $5M for lending modernization. Nothing public. System sees intent = 0. Reality: intent = 100.*

FAIL-DARK-001
"Dark Intent Detection Framework" — 4 source categories: (1) Relationship Intelligence: manual entry by BD, confidence 0.80 (anonymous tips 0.50). (2) Conference Intelligence: manual with event/speaker/quote, confidence 0.75 (specific quotes 0.90). (3) Ecosystem/Subsidiary Signals: automated detection + manual linkage, confidence 0.65. (4) First-Party Behavioral: website analytics from identified contacts, confidence 0.70 (anonymous 0.40).
Rationale: *Dark intent can only be detected through human networks and behavioral signals, not public news.*

FAIL-DARK-002
Relationship intelligence gets FULL scoring weight (same as P1 ACTIVE_PROCUREMENT) BUT caps ICS Verification at 50 unless corroborating public signal arrives within 30 days.
Rationale: *Connector who says "they're buying" is usually right, but we can't independently verify a private conversation. High priority, moderate confidence.*

FAIL-DARK-003
"Manual Intelligence Incentive System": (a) every manual signal timestamped and attributed, (b) weekly dashboard showing entries per BD member and conversion rate, (c) attribution credit when manual signal leads to meeting within 60 days, (d) if total team entries <5/month, alert sales leadership, (e) post-conference/call Slack prompt within 24h: "Any intelligence to log?"
Rationale: *If humans forget to log dark intent, the system is blind.*

FAIL-DARK-004
"Connector Intelligence Protocol": (a) designated Decimal relationship manager per connector org, (b) monthly structured check-in: "Any lending/origination/onboarding initiatives at your Saudi clients?", (c) connector intelligence entered as CONNECTOR:[name] source, (d) +0.10 confidence boost over standard manual signals, (e) track connector reliability over time.
Rationale: *Connectors (ITC, Backbase, Tarabut, Geidea, KPMG, Mastercard) are Decimal's eyes inside banks.*

FAIL-DARK-005
"Post-Mortem: Lost Before We Knew" — When system discovers a bank selected a competitor for a deal never detected: log as MISSED_OPPORTUNITY with account, competitor, category, value, how_discovered. Root cause categories: No_Public_Signals / Signals_Missed / Detected_Not_Acted / Relationship_Gap. Quarterly review: how many missed? Estimated revenue lost? Most common root cause?
Rationale: *You can't fix what you don't measure. Missed opportunities are the most expensive failures. — 1I. Failure Mode 8: Signal Conflict (4 Rules) Multiple signals contradict each other: New CTO hired (positive) + Competitor signed (negative) + Funding round (positive) + Layoffs (negative). Current engine has no conflict resolution.*

FAIL-CONFLICT-001
3 conflict types: DIRECT_CONTRADICTION (same topic, opposite conclusions — most recent wins by default, flag for human verification), MIXED_SIGNALS (both opportunity and risk simultaneously — not a contradiction, AI must reason about Decimal-specific relevance), TEMPORAL_SHIFT (earlier signal suggested opportunity, later signal closed it — later signal takes priority).
Rationale: *Different conflict types require different resolution strategies.*

FAIL-CONFLICT-002
"Signal Arbitration" for MIXED_SIGNALS: Step 1: Tag each signal by Decimal product relevance (ORIGINATION/LENDING/ONBOARDING/WORKFLOW/OPEN_BANKING/GENERAL). Step 2: If negative signal specifically names competitor winning in Decimal's category = "BLOCKED." If competitor won different category = potential complementary opportunity. Step 3: Net positive signals >= 1 in Decimal's category = proceed. Net negative = "CONFLICTED_ACCOUNT" for human review.
Rationale: *A competitor winning core banking doesn't block Decimal's origination play.*

FAIL-CONFLICT-003
AI reasoning chain must acknowledge ALL signals for conflicted accounts. Agent 4 (Pain) must reason about how conflicting signals create different pains. Agent 5 (Strategy) must choose angle that acknowledges complexity. Agent 7 (QC) must verify: "Does this message reflect the FULL signal picture?" Cherry-picked messaging = QC REJECTS.
Rationale: *Pretending the competitor deal doesn't exist is worse than addressing it.*

FAIL-BIAS-001
"Outcome Attribution Framework" — When sequence completes without meeting, classify into: SIGNAL_MISS (signal was wrong), TIMING_MISS (too early/late), CONTACT_MISS (wrong person), MESSAGE_MISS (right everything, message didn't resonate), DELIVERY_MISS (never reached contact), EXTERNAL_BLOCK (everything right, external factor prevented response), UNKNOWN. Each feeds its OWN feedback loop. SIGNAL_MISS -> signal weights. CONTACT_MISS -> persona targeting. MESSAGE_MISS -> prompts. DELIVERY_MISS -> infrastructure. EXTERNAL_BLOCK -> NO negative feedback to any engine.
Rationale: *Each failure type has a different cause and solution. Contaminating feedback loops produces wrong learnings.*

FAIL-BIAS-003
"Success Attribution Honesty" — When meeting IS booked, classify
Rationale: *SIGNAL_DRIVEN (signal was the reason), RELATIONSHIP_DRIVEN (warm intro was primary driver), TIMING_LUCK (contacted during existing evaluation by coincidence), PERSISTENCE_DRIVEN (no response to touches 1-3, responded to touch 4). WHY: Prevents the system from taking credit for luck and over-optimizing for non-causal factors.*

FAIL-HUMAN-002
"Human Resilience Principles": (a) No single point of human failure — every role has a backup, (b) escalation timers on EVERY touchpoint, (c) graceful degradation — if human fails SLA, system escalates/queues/proceeds with guardrails, doesn't stop, (d) "Human Health Score" — % of SLAs met across all touchpoints weekly. Target >= 85%. Below 70% = operational review.
Rationale: *The system must be designed to survive any single human failure.*

FAIL-HUMAN-003
"Reviewer Pool Architecture": Minimum 2 trained reviewers always. Training: 3 days (Day 1 = understanding AI chain, Day 2 = practicing with historical examples, Day 3 = live review with supervision). Certification: first 20 reviews audited, >= 90% agreement = certified. Vacation protocol: no reviewer goes on leave without backup confirmed active.
Rationale: *Reviewer quality directly determines outreach quality.*

FAIL-HUMAN-005
"Bottleneck Early Warning System": Leading indicators: review queue depth >10 = AMBER, >20 = RED + activate backup. Average review time trending 50%+ above 30-day average = reviewer fatigued. Reviewer no reviews in 6+ hours during business hours = availability check. Manual signal entries declining 2+ months = intelligence flow drying up. Lagging indicators: messages expired from queue, Golden Windows missed, handoff SLA breaches.
Rationale: *Detecting bottlenecks early prevents pipeline stalls. — 1L. Module 0: Intelligence Governance Engine (11 Rules) Module 0 sits ABOVE all other modules. Every other module asks 'What should we DO?' Module 0 asks 'Can we TRUST what we think we know?' This is the meta-layer that prevents the system from acting...*

FAIL-MISATTR-002
Multi-Entity Signals: When signal mentions 2+ accounts, create ONE record per account with shared group_id. Signal type determined independently per account context. Partner mentions stored as metadata, not separate signals.
Rationale: *"SNB and Riyad Bank announce joint initiative" creates two signals, one per bank.*

FAIL-MISATTR-003
Misattribution Recovery: Re-map to correct account, recalculate BOTH accounts, check if outreach referenced the signal — if message said "your digital initiative" and it was actually the other bank's initiative = reputation-critical error, immediate human intervention. Log as ENTITY_RESOLUTION_ERROR, update aliases.
Rationale: *Sending SNB's initiative to Riyad Bank's CTO is permanently trust-destroying.*

FAIL-MISATTR-004
Pre-Outreach Entity Verification: QC CHECK ZERO (before all other QC checks): company name in message matches account name, signal referenced belongs to this account, no MULTI_ENTITY flags. Fail = QC REJECTS with "ENTITY_VERIFICATION_FAILED." Hard block, no override without human confirmation.
Rationale: *This is the most expensive possible error.*

FAIL-SCOPE-001
Account Relevance Scope: Every account has defined relevance_scope — business divisions Decimal is relevant to. Example: SNB = ["Retail Banking", "SME Lending", "Digital Banking", "Open Banking"]. Out-of-scope signals stored but EXCLUDED from scoring.
Rationale: *SNB's insurance technology investment shouldn't inflate scoring for Decimal's lending pitch.*

FAIL-SCOPE-002
Contact Relevance Filter: Only contacts in scope divisions join active buying committee. "Head of Treasury Technology" at a retail-banking-scoped account = stored but OUT_OF_SCOPE for outreach. Exception: C-suite always in scope.
Rationale: *Treasury technology contacts receiving lending platform outreach is irrelevant and unprofessional.*

FAIL-SCOPE-003
Scope Expansion Protocol: Changed only through: (a) manual update with documented reason, (b) product launch batch-update for all applicable accounts. Changes logged, trigger: re-evaluation of OUT_OF_SCOPE signals/contacts, score recalculation.
Rationale: *Scope expansion is strategic, not automatic.*

FAIL-OWNER-001
Owner Load Limits: Max 10 HOT accounts per owner (require active management). Max 20 WARM. Owner at capacity + new HOT account = assign to next available. All owners at capacity = "CAPACITY_OVERFLOW" alert, max 48h wait for HOT accounts.
Rationale: *Overloaded owners become bottlenecks across all their accounts.*

FAIL-OWNER-002
Owner Transition Protocol: Owner leaves/reassigned → 24h TRANSITION_HOLD (no new outreach, active sequences continue). Leadership reassigns within 24h based on expertise + load + relationship continuity. New owner gets Transition Brief per account. ENGAGED contacts flagged: "ACTIVE CONVERSATION — review immediately."
Rationale: *Orphaned accounts with active conversations lose prospects.*

FAIL-OWNER-003
Owner Override Rules: CAN: delay outreach up to 14 days with reason, reprioritize contacts, edit messages, add manual signals. CANNOT: permanently suppress HOT account without leadership approval, change scoring formula, override QC rejections, delete records. All overrides logged with type, reason, duration, approved_by.
Rationale: *Owners need flexibility — but not unchecked power.*

FAIL-FATIGUE-001
Account Fatigue Score = (completed sequences with no positive reply) x 10 over rolling 12 months. 0-20: normal. 30-40: CAUTION, next outreach requires owner justification. 50+: FATIGUED, 180-day reset, re-engagement only through different channel or connector introduction. Score resets to 0 only after successful engagement.
Rationale: *Repeated unsuccessful outreach annoys prospects and damages reputation.*

FAIL-FATIGUE-002
Contact Fatigue: Max 3 lifetime sequences per contact. After 3 with no positive reply = "OUTREACH_EXHAUSTED," permanently removed from outreach pool. Exception: contact initiates contact with Decimal (inbound) = fatigue cleared, one new sequence allowed.
Rationale: *Enterprise professionals talk. A 4th campaign will create active anti-Decimal sentiment.*

FAIL-FATIGUE-003
Outreach Freshness Requirement: New sequence requires at least ONE of: (a) NEW signal since last sequence, (b) DIFFERENT contact targeted, (c) DIFFERENT outreach angle (AI confirms substantially different), (d) connector warm introduction. None met = sequence blocked.
Rationale: *"We have no new reason to reach out" means don't reach out.*

FAIL-FATIGUE-004
Nurture Fatigue: WARM accounts max 1 nurture touch/month. 6 consecutive zero-engagement touches = "NURTURE_EXHAUSTED," removed from nurture. Re-enters only if promoted to HOT then back to WARM.
Rationale: *Monthly emails to someone who never opens is spam with extra steps.*

FAIL-DECAY-001
Data Decay Rates: Contact role/title: 12-18 months. Email: 6-12 months (re-verify before each sequence). Strategic priorities: 3-6 months. Digital maturity: 6-12 months. Technology stack: 6-12 months. Buying committee composition: 3-6 months. Competitor landscape: 3-6 months.
Rationale: *KSA banking executives change roles every 1-3 years. Data without refresh becomes fiction.*

FAIL-DECAY-003
Data Staleness Threshold: If most recent enrichment >120 days AND no new signal in 60 days: ICS Data Completeness decays to 50%, Verification decays to 40%. Combined effect: ICS drops significantly, potentially triggering GOV-001 investigate-only gate.
Rationale: *System says "we used to know about this account, but our knowledge is too old to trust." — 2L. Reply-Based Account Pause Logic (4 Rules)*

FAIL-RUBBER-001
"Anti-Rubber-Stamping Mechanisms": (a) Minimum review time tracking: <30 seconds for email or <15 for LinkedIn = "SPEED_REVIEW" flag (meaningful review takes 60-90 seconds). (b) Spot-check canaries: weekly, system inserts deliberately flawed message (wrong company name, non-existent signal). Approved canary = "CANARY_MISSED" alert. Flaw caught before sending. Reviewer notified. (c) Review Quality Score per reviewer: canary catches, edit rate, time per review, agreement with post-hoc audits. (d) Rotation: reviewers don't review same account repeatedly. Monthly rotation.
Rationale: *After 50 good messages, reviewer assumes #51 is fine. Canaries test vigilance.*

FAIL-RUBBER-002
"Review Quality Culture": Canary system framed as QUALITY TOOL not surveillance: (a) all reviewers told upfront during training, (b) results private, not shared with team, (c) missed canary = coaching not discipline, (d) Review Quality Score for training needs, not performance evaluation, (e) canary frequency decreases as reviewer quality improves.
Rationale: *Surveillance-framed systems create resentment and gaming. Quality-framed systems create improvement. FM2 — AI Hallucination (2 Rules)*

FAIL-HALLUC-001
"Three-Layer Hallucination Prevention": Layer 1 — Generation Constraint: Agent 6 explicitly instructed to use ONLY facts from enrichment, signals, and Approved Claims Library. Unverifiable = "CLAIM_UNVERIFIED" placeholder. Layer 2 — Automated Fact-Check: before Agent 7, cross-reference every entity name, number, date, claim against databases. Unmatched claims flagged with specific text highlighted. Layer 3 — Human Review: flagged claims shown in RED. Unflagged in normal text. Visual hierarchy focuses human attention on highest-risk parts.
Rationale: *Triple-layer means hallucination must fool generator, fact-checker, AND human.*

FAIL-HALLUC-002
"Hallucination Incident Response": If hallucinated claim IS sent: Step 1: account owner notified within 1 hour. Step 2: assess — did prospect notice? Step 3: if noticed, owner contacts personally, acknowledges error honestly, provides correct info. Step 4: incident logged as "HALLUCINATION_BREACH" with which layer(s) failed + root cause. Step 5: add hallucination pattern to fact-checker negative rules.
Rationale: *Honest correction builds more trust than silence. FM3 — AI Staleness (2 Rules)*

FAIL-STALE-001
"AI Currency Maintenance Schedule": Approved Claims Library: monthly (triggered by product launch, client win, certification, pricing change). Persona Prompts: quarterly (triggered by persona mismatch feedback). Strategy Prompts: monthly (triggered by market/competitive changes). Cultural Rules: quarterly (triggered by policy changes, lessons learned). Competitive Intelligence: continuously. Signal Classification: monthly (triggered by false positive thresholds). QC Criteria: quarterly (triggered by systematic failures). All version-controlled. Rollback within 1 hour if degradation.
Rationale: *AI that doesn't evolve becomes irrelevant as markets change.*

FAIL-STALE-002
"AI Performance Degradation Detection": Track monthly — thresholds for "STALE_AI" trigger: QC approval rate drops 10+ points from 90-day average, human override rate increases 10+ points, reply rate declines 30%+ (controlling for seasonality), average edit distance increasing, MAJOR_REWRITE rate >20% for any category. STALE_AI triggered: emergency prompt review, enrichment audit, competitive check, claims library verification. Fix within 7 business days.
Rationale: *Gradual degradation is invisible reviewing individual messages. Systematic measurement catches decline. FM4 — Human Over-Reliance (1 Rule)*

FAIL-OVERRELY-001
"Human Independent Judgment Preservation": (a) Monthly "Red Team": owners independently rank top 5 accounts WITHOUT seeing system ranking first, then compare. Divergence is GOOD. Perfect alignment after 6+ months = possible over-reliance. (b) AI Confidence Display: always show confidence level with recommendations. Never present as certainties. (c) Override encouragement: if owner NEVER overrides in 90 days, manager conversation: genuinely agreeing or defaulting? (d) Market Sensing Reports: monthly BD team report based on THEIR conversations, independent of AI analysis. Compare "what market tells us" vs "what AI tells us."
Rationale: *An AI-dependent team cannot function during an outage. FM5 — Wrong Metric Optimization (2 Rules)*

FAIL-WRONGMETRIC-001
"Metric Hierarchy": NORTH STAR: pipeline revenue from ABM-sourced opportunities. PRIMARY: meetings booked, positive reply rate. SECONDARY: total reply rate, signal-to-meeting conversion. OPERATIONAL: open rate, click rate, LinkedIn accept rate, QC pass rate. AI must NEVER optimize operational at expense of primary. A change that increases open rates but decreases positive reply rates is BAD. Feedback loops ONLY recalibrate based on primary/north star. Operational metrics for diagnostics only.
Rationale: *Optimizing vanity metrics produces impressive dashboards and empty pipeline.*

FAIL-WRONGMETRIC-002
"Metric Manipulation Detection": Divergence between operational and primary = warning: open rate up but reply flat = misleading subjects. QC pass up but override also up = QC weakened. LinkedIn accept up but DM reply flat = connection good, follow-up bad. Monthly check required.
Rationale: *Metrics that improve independently of outcomes indicate gaming or miscalibration. FM6 — AI Memory Leakage (2 Rules)*

FAIL-LEAK-001
"Account Isolation Principle": Each reasoning chain receives ONLY that account's data: enrichment, signals, contact personas, plus universal Approved Claims Library and general strategy guidelines. Must NOT receive other accounts' data, engagement outcomes, or response patterns.
Rationale: *What works at SNB may fail catastrophically at Al Rajhi. Each bank has its own culture, priorities, and politics.*

FAIL-LEAK-002
"Cross-Account Learning Rules": Patterns CAN be learned across accounts if: (a) derived from 5+ accounts, (b) about PERSONA TYPES not specific people, (c) about MESSAGE CHARACTERISTICS not specific content, (d) applied as TENDENCY not hard rule. ALLOWED: "CTOs tend to prefer messages under 100 words." NOT ALLOWED: "CTO at Bank A liked compliance pitch — use for Bank B."
Rationale: *Aggregate behavioral patterns are useful. Specific account strategies are not transferable. DEEP DRILL: Trust Progression Model (1 Rule)*

HUMAN Rules (4)

HUMAN-AUTO-001
Three-phase human review reduction: Phase 1 (Days 0-90): 100% human review. Phase 2 (91-180): Optional for P3/P4 + non-C-suite if QC rate >85% and override 92% and override <5%. C-suite messages: ALWAYS human reviewed, permanently.
Rationale: *C-suite errors are reputation-destroying. Lower-risk messages can be automated once trust is established.*

HUMAN-AUTO-002
Escalation chain for message review: Hour 0-4: Primary reviewer (account_owner). Hour 4-8: Backup reviewer (buddy). Hour 8-24: Team lead. Hour 24+: Message expires, must be regenerated with fresh context.
Rationale: *Context goes stale. "This week's announcement" becomes last week's after 48 hours.*

HUMAN-AUTO-003
Track edit distance on human edits: MINOR_TWEAK 50%. If MAJOR_REWRITE rate exceeds 30% for a signal_type or persona_type, trigger prompt optimization review.
Rationale: *High rewrite rates mean the AI is failing for that category — fix the prompts, not the individual messages.*

HUMAN-AUTO-004
Human-added factual claims must be logged separately from AI-generated claims.
Rationale: *The system validates AI claims against enrichment data. Human additions are not validated but must be tracked for audit purposes. Timing Windows & Volume Controls*

HUMANREVIEW Rules (5)

HUMANREVIEW-001
"Accountability Attachment Record": Every outbound action stores: AI_RECOMMENDED_BY (chain version), REVIEWED_BY (human name + timestamp), ACCOUNT_OWNER (ultimately responsible human), DECISION (approve/edit/reject), REASONING (linked to HUMANDEC-001 taxonomy). Record is IMMUTABLE — cannot be edited after the fact. Answers: "When this message was sent to SNB's CEO, who approved it and why?"
Rationale: *AI cannot appear in a meeting to explain why it sent a message. A human can. Accountability attaches to persons, not systems.*

HUMANREVIEW-002
"Strategic vs Execution Review": STRATEGIC (higher stakes, less frequent): covers account targeting, stakeholder selection, outreach narrative, competitive positioning, timing strategy. Reviewer: owner or BD leadership (Level 2-3). Frequency: once per account per outreach cycle. EXECUTION (lower stakes, more frequent): individual message quality, wording, CTA. Reviewer: any certified (Level 1+). Frequency: per message during MVP, reducible per trust. CRITICAL: Strategic review is NEVER automated — even at FULL PARTNERSHIP, human always approves account STRATEGY. Execution review within approved strategic boundaries can auto-send if trust permits.
Rationale: *Getting strategy wrong means EVERY message is wrong. This is the high-leverage review.*

HUMANREVIEW-003
"Reasoning Checkpoint Access": Reviewer interface allows reviewing ANY intermediate step: expandable Agent 1 → 2 → 3 → 4 → 5 → 6 → 7. DEFAULT view: final message + strategy summary + signal reference (fast routine review). EXPANDED view: full chain (when something feels off). MANDATORY expansion: C-suite contact, SIGNAL_CONFLICT flag, ICS <60, first message to new account, Verifiability Score <50.
Rationale: *Reviewing only the final message is reviewing the symptom. Reviewing reasoning is reviewing the cause.*

HUMANREVIEW-004
"Reviewer Competency Tiers": Level 1 OPERATIONAL: P3/P4 to Director and below. Requires 4-week onboarding + certification (20 audited reviews, >=90% agreement). Level 2 STRATEGIC: P1/P2, VP contacts, strategic reviews, conflicted accounts. Requires Level 1 experience (100+ reviews) + demonstrated strategic judgment + BD experience. Level 3 EXECUTIVE: C-suite messages, Domino Account outreach, market strategy, override approvals. Requires senior BD experience + deep KSA banking knowledge + multi-account relationship context. Messages routed to appropriate level. C-suite WAITS for Level 3 — better to delay than have junior reviewer approve CEO message.
Rationale: *Not all reviews are equal. Not all reviewers are qualified for all reviews.*

HUMANREVIEW-005
"Review Step Sunset Assessment": Every 6 months, per message category (signal priority x contact seniority x trust level): (a) VALUE_ADDED rate from HUMANDEC-005, (b) BLOCKED_HARM rate, (c) cost of this review step. VALUE_ADDED >20% OR BLOCKED_HARM >2% → JUSTIFIED, keep. VALUE_ADDED 5-20% AND BLOCKED_HARM <2% → QUESTIONABLE, consider spot-check (30% random). VALUE_ADDED <5% AND BLOCKED_HARM <1% → NOT JUSTIFIED, candidate for removal per trust progression. Exception: ANY category involving C-suite or P1 → ALWAYS justified regardless of metrics.
Rationale: *Review that never changes outcomes is overhead. But some categories must stay reviewed because the consequence of the ONE error is too high. ITERATION 8 — WHY AI REASONS THE WAY IT DOES Good outcome does not equal good reasoning. Bad outcome does not equal bad reasoning. An AI that produces a meeting through lucky reasoning will produce a dis...*

ICS Rules (7)

ICS-CALC-001
ICS (0-100) calculated from 6 weighted dimensions: Signal Diversity 20%, Signal Recency 15%, Data Completeness 20%, Contact Coverage 15%, Verification Level 15%, Source Reliability 15%.
Rationale: *Intelligence reliability is multi-dimensional. Strong in one area doesn't compensate for weakness in another.*

ICS-DIV-001
Signal Diversity scoring: 0 sources = 0pts. 1 source = 20pts (single source = unreliable). 2 sources = 50pts. 3 sources = 75pts. 4+ sources = 100pts.
Rationale: *Multi-source triangulation is fundamentally more reliable than single-source intelligence.*

ICS-REC-001
Signal Recency scoring: within 7 days = 100pts. 8-21 days = 70pts. 22-45 days = 40pts. 46-90 days = 15pts. No signal or 90+ days = 0pts.
Rationale: *Old intelligence is unreliable. A bank's priorities can shift in weeks.*

ICS-COMP-001
Data Completeness checklist (each item adds points): Company name+segment+country: +10. AI account summary: +15. Digital maturity assessed: +10. Open banking readiness: +10. Strategic initiatives (>=2): +15. Technology stack known: +10. Procurement complexity: +10. Competitive landscape: +10. Recent news within 30 days: +10. Total: 100.
Rationale: *Every missing data point is a gap the AI has to guess around.*

ICS-CONTACT-001
Contact Coverage: 0 contacts = 0pts. 1 = 15. 2 = 30. 3 = 50. 4-5 spanning 2+ departments = 75. 6+ spanning 3+ departments including C-level = 100.
Rationale: *Enterprise deals require multi-threaded engagement. Single contact = single point of failure.*

ICS-VERIF-001
Verification from: verified emails (up to 40pts), signals from high-reliability sources (up to 30pts), enrichment from premium sources (up to 30pts). Formula: (verified_email% x 0.4 + verified_signal% x 0.3 + premium_enrichment% x 0.3) x 100.
Rationale: *Unverified data is unreliable data.*

ICS-DECAY-001
ICS decays by 2 points per week after last enrichment or signal ingestion if no new data arrives.
Rationale: *Stale confidence is false confidence. Even if nothing has been contradicted, understanding degrades over time.*

LAW2 Rules (2)

LAW2-EXCEPTION-001
Decision Complexity Classification: SINGLE_DECISION_MAKER (<100 employees, CEO decides) = 1 contact sufficient. SMALL_COMMITTEE (100-500, 2-3 deciders) = 2 contacts min. STANDARD_COMMITTEE (500-5000, 5-7 deciders) = standard rules. COMPLEX_COMMITTEE (5000+, 10+ stakeholders, procurement-driven) = 3+ contacts, human-led early.
Rationale: *Treating a 30-person fintech like a 30,000-person bank wastes resources.*

LAW2-EXCEPTION-002
Lightweight Pipeline for SINGLE_DECISION_MAKER: abbreviated enrichment (CEO only), abbreviated AI chain (skip persona psychology + multi-angle strategy), shorter sequence (3 touches), binary reachability (can reach CEO or can't).
Rationale: *Speed matters more for startups. CEO who raised funding makes vendor decisions in 30 days.*

LAW4 Rules (1)

LAW4-EXCEPTION-002
2+ personal signals on same topic from different individuals within 30 days = auto-upgrade to ORGANIZATIONAL.
Rationale: *Multiple people independently signaling = organizational-level activity.*

LAW5 Rules (2)

LAW5-EXCEPTION-001
Contact Engagement Heatmap: Per-contact engagement breakdown maintained alongside account-level score. Shows which contacts most responsive. Feeds human strategy and persona performance analytics. Does NOT independently trigger state changes.
Rationale: *Contact data INFORMS humans but doesn't drive system actions.*

LAW5-EXCEPTION-002
Contact Velocity Alert: Single contact with rapidly escalating engagement (open -> click -> website -> reply in 72h) creates CONTACT_VELOCITY_ALERT for account_owner. Does NOT change account score/state.
Rationale: *Operational signal for human to prioritize.*

LAW6 Rules (1)

LAW6-EXCEPTION-001
Contact-Focal Moments: Temporary zoom-in events when one contact becomes primary unit: meeting booked (AI brief focuses on this person), referral received (outreach crafted for referred person), connector introduction (messaging optimized for this person), procurement RFP (response formatted for procurement contact). Temporary, not architectural.
Rationale: *Strategy is account-level. Execution has moments when one person is the game.*

QUEUE Rules (1)

QUEUE-ARCH-001
CONFLICT RESOLUTION (FIN-C05): P1 processing (scoring, enrichment, message generation, QC) is immediate and runs 24/7 regardless of business hours. Human review and SEND are decoupled from processing: a QC-passed message enters the Ready-to-Fire Queue and waits for (a) human review per the active Trust level and (b) the next valid KSA send window. A P1 signal at 11pm Thursday is fully processed by ~1am, reviewed Sunday morning, and sent in Sunday 9am-12pm. EVENT-RESP-001 and HUMAN-AUTO-002 both defer to this queue. Rationale: encodes GOVERNANCE-004 so immediate-processing and mandatory-review no longer collide.

TEMPORAL Rules (4)

TEMPORAL-DECISION-001
"Timing Paradox Rule": For timing-sensitive decisions calculate: COST_OF_ACTING_NOW (risk of premature × impact of TOO_EARLY) vs COST_OF_WAITING (risk of missing window × impact of TOO_LATE + daily information decay). Waiting > Acting → ACT NOW, accept uncertainty. Acting > Waiting → WAIT with maximum wait period. P1: cost of waiting very high, default ACT NOW. P2: can wait 48-72h for better intelligence. P3/P4: can wait for full enrichment.

TEMPORAL-DECISION-002
"Window State Confidence": HIGH >=80% (multiple corroborating signals, act on it). MODERATE 50-79% (some signals, act with lighter touch — positioning or exploratory). LOW <50% (ambiguous, do NOT launch full outreach, investigate first). Display confidence alongside state: "Window: OPENING (62%). Basis: new CTO + Q1 timing. Missing: budget confirmation."

TEMPORAL-DECISION-003
"Cross-Account Timing Coordination": Peer group / shared board accounts: (a) sequence strategically — stronger opportunity first, positive signal becomes social proof for second. (b) Domino Account timing takes priority. (c) NEVER same-signal outreach to two connected accounts on same day — stagger 3+ business days. (d) Referral from Account A to Account B: approach B within 48h while endorsement is fresh.

TEMPORAL-DECISION-004
"Internal Readiness Gates": External timing (window OPEN) must align with internal readiness across 6 dimensions: enrichment sufficient (ICS >=40), message quality (QC passed), reviewer available, account owner assigned, competitive intelligence known, relevant collateral exists. Each not-ready has specific response: emergency enrichment, fix message, queue for reviewer, assign owner, flag blind spot, proceed anyway for P1/P2 but prepare collateral in parallel.

TRUST Rules (2)

TRUST-001
"Trust Progression Framework" — 6 levels: ZERO TRUST (Day 0): AI drafts only, cannot queue. 100% review. Advance after 30 days no incidents. INITIAL TRUST (Day 30+): AI queues for review. Auto-enrichment, auto-scoring. QC >60% to advance. DEVELOPING TRUST (Day 60+): QC >80%. Review SLA can extend to 8h for P3/P4. ESTABLISHED TRUST (Day 90+): QC >85%, override 92%, override <5%. P2 Director optional review. C-suite ALWAYS human. FULL PARTNERSHIP (Day 360+): AI handles routine. Humans focus on relationships/strategy. C-suite/P1 always human. Quarterly full audit. CRITICAL: Trust can REGRESS. Single hallucination = drop one level. Reputation damage = drop to INITIAL.
Rationale: *Trust earned incrementally, lost instantly — mirrors human trust in enterprise relationships. ITERATION 1 — AI LEARNING & EVOLUTION ARCHITECTURE How does the AI actually get smarter without getting dumber? Learning in a h...*

TRUST-AUTHORITY-001
CONFLICT RESOLUTION (FIN-C01): The Trust Progression Framework (TRUST-001) is the single source of truth for autonomy and review-reduction. The day-based phases in HUMAN-AUTO-001 and FAIL-HUMAN-004 are minimums that cannot fire before the corresponding Trust level is reached. Where the three rules disagree on timing or gate, the LATER date and the STRICTER gate always win. Rationale: three independent schedules for the same transition created ambiguity; a single authority with conservative tie-breaking removes it.

Section 2: Master Entity Model (56 rules)

ACCT Rules (1)

ACCT-STATE-001
Account maintains a denormalized current_state block (tier, window_state, readiness, stage, last_decision_ts) updated by the scoring engine on every recalculation. Read-optimized cache; authoritative history lives in the intelligence entities.
Rationale: *#4 — Why does that denormalization not create an integrity risk? Because the cache is write-controlled: only the scoring engine writes current_state, and every write emits a System_Event. If cache and source disagree, the event log reconstructs truth. The Account never accepts a direct human edit to current_state.*

CONFLICT Rules (3)

CONFLICT-RECORD-001
Each rule conflict is recorded with the conflicting rules, context, the tier-hierarchy resolution applied, and the outcome. Silent resolution is prohibited.
Rationale: *#2 — Why resolve via the 6-tier hierarchy rather than ad hoc? Ad hoc resolution is unpredictable and unauditable. The Constraint Hierarchy (Legal > Safety > Relationship > Timing > Revenue > Optimization) gives a stable, explainable ordering.*

CONFLICT-HIERARCHY-001
Conflicts resolve by the 6-tier Constraint Hierarchy. The winning tier and rule are recorded. Conflicts unresolvable within time limits (4h/24h) escalate to human review.
Rationale: *#3 — Why build a precedent library from resolutions? Repeated identical conflicts shouldn't be re-litigated. After enough consistent resolutions, the pattern becomes a standing rule, speeding future decisions.*

CONFLICT-TIMEBOX-001
Conflicts must resolve within tiered limits (4h automated / 24h with human). Breaching the limit auto-escalates; timing-critical conflicts get the shortest limit.
Rationale: *#5 — Deepest reason this entity exists? It makes the system's value hierarchy explicit and consistent — ensuring that when principles collide, safety and relationships reliably outrank short-term revenue. It exists so the system's character is stable under pressure.*

CONTACT Rules (3)

CONTACT-ROLE-001
Contact stores identity and persona; deal-specific positioning lives on Buying_Committee_Member. A Contact with no active deal positioning is still valid.
Rationale: *#4 — Why must persona live on Contact, not be inferred per message? Persona (motivations, risk posture, decision style) is stable and expensive to derive. Re-inferring per message wastes capital and breeds inconsistency. Storing it makes reasoning Step 4 deterministic and auditable.*

CONTACT-PERSONA-001
Contact stores structured persona_profile (role_motivations, decision_style, risk_posture, communication_preference, evidence_orientation) with provenance + confidence. Stale personas (>180 days or post-role-change) flagged for re-enrichment.
Rationale: *#5 — Deepest reason Contact exists? Trust is built person-to-person; the system must never treat a human as a disposable address. Contact embodies the Relationship and Market Memory principles — forcing the system to remember who it spoke to and what it owes them.*

CONTACT-GOVERNANCE-001
No outreach may be generated to a Contact without checking interaction_lineage for prior negative sentiment, do-not-contact flags, or active fatigue. A burned relationship is a hard gate, not a soft penalty. CONTACT — Field Specification

ENGAGE Rules (4)

ENGAGE-SOURCE-001
Engagement_Event records recipient-side or third-party-observed actions, distinct from outbound Messages. Each records actor, action_type, and source-of-observation.
Rationale: *#2 — Why feed scoring in near-real-time? Engagement is a signal — an inbound buying-intent indicator. A reply or repeated profile views should retrigger scoring (Principle 1).*

ENGAGE-TRIGGER-001
High-intent Engagement_Events (reply, meeting-accept, repeated content interaction) retrigger account scoring and may open/widen a Buying_Window. Engagement is a first-class signal class.

ENGAGE-GRANULARITY-001
Engagement is stored as discrete timestamped events, never pre-aggregated. Aggregations computed at read time so velocity, clustering, and recency remain recoverable.
Rationale: *#4 — Why link to the specific Message that provoked it? Attribution requires the cause-effect pair. To learn which message earned the reply, the event must point back to the triggering Message and its Reasoning_Chain.*

ENGAGE-LEARNING-001
Every Engagement_Event contributes to a Feedback_Loop_Entry. Absence of engagement after N touches is itself a learning signal (triggers sequence pause / strategy review). ENGAGEMENT_EVENT — Field Specification Field Type Business Rule Source event_id UUID Immutable ENGAGE-SOUR CE-001 account_id / contact_id UUID FKs — action_type Enum OPEN / CLICK / REPLY / PROFILE_VIEW / ENGAGE-SOUR CONTENT_INTERACTION / MEETING_ACCEPT CE-001 triggering_message_id UUID FK to Message (if traceable) ENGAGE-ATTRI BUTION-001 observed_via Enum Source of observation ENGAGE-SOUR CE-001 intent_class Enum HIGH / MEDIUM / LOW (drives retrigger) ENGAGE-TRIG GER-001 occurred_at Timestamp Discrete, never aggregated ENGAGE-GRAN ULARITY-001 TIER 2 — INTELLIGENCE ENTITIE...

ENRICH Rules (3)

ENRICH-COST-001
Enrichment records source_cost and counts against the account's capital budget. Enrichment of accounts below a fit threshold requires justification; speculative enrichment of COLD accounts is rate-limited.
Rationale: *#3 — Why store freshness/expiry on enrichment? Stale enrichment is a named failure mode. Firmographics, headcount, and roles change; acting on year-old data damages credibility. Each fact needs a confidence-decay and re-fetch trigger.*

ENRICH-FRESHNESS-001

Each enriched fact carries source_date and a type-specific staleness window. Stale facts are flagged and must be re-verified before use in reasoning; staleness lowers Intelligence Confidence.

Rationale: *#4 — Why reconcile conflicting sources rather than overwrite? Apollo and Clearbit may disagree. Last-write-wins silently destroys signal. The system must record both and resolve by source-reliability, preserving the conflict for audit.*

ENRICH-RECONCILE-001

When sources conflict, both values are retained with source-reliability weighting; the resolved value records its rationale. Unresolved high-stakes conflicts lower confidence and may trigger human review.

Rationale: *#5 — Deepest reason Enrichment exists? It is how the system converts the unknown into the known — the sensory apparatus that turns a bare account name into an actionable intelligence profile. Without provenance and freshness, knowledge silently rots.*

INIT Rules (5)

INIT-RECORD-001

Each known Initiative is stored with type, sponsor, maturity stage, and criticality. Outreach reasoning must map Decimal's value to a specific initiative where one exists.

Rationale: *#2 — Why track initiative maturity stage? A just-announced initiative buys differently from one mid-build. Maturity determines whether Decimal is early (shaping) or late (displacing an incumbent), which changes the entire strategy.*

INIT-MATURITY-001

Initiative maturity (ANNOUNCED/PLANNING/BUILDING/LIVE/SCALING) is tracked. Project-maturity models set the engagement angle; entering at BUILDING vs LIVE triggers different playbooks.

Rationale: *#3 — Why track initiative criticality? Not all initiatives are equal. A CEO-sponsored transformation outranks a departmental experiment. Criticality drives how much capital the opportunity justifies.*

INIT-CRITICALITY-001

Each Initiative carries a criticality rating (strategic weight, executive sponsorship, budget size). Criticality feeds the opportunity equation; low-criticality initiatives cap allocated capital.

Rationale: *#4 — Why link initiatives to sponsors and windows? An initiative without a sponsor is a rumor; an initiative whose sponsor just left is dying. Linking sponsor (a committee member) and any buying window keeps initiative intelligence live, not static.*

INIT-SPONSOR-001

Each Initiative links its executive sponsor (Buying_Committee_Member) and any associated Buying_Window. Sponsor departure triggers re-assessment of initiative viability.

Rationale: *#5 — Deepest reason Initiative exists? It is the bridge between Decimal's capability and the account's actual spending. It exists because money flows to initiatives — selling to an account without an initiative is selling to no one.*

INIT-PRIMACY-001

Where no initiative can be identified, the account is treated as lower-readiness and outreach shifts to discovery/education rather than solution-selling. INITIATIVE — Field Specification Field Type Business Rule Source initiative_id UUID Immutable INIT-RECORD-0 01 account_id UUID FK to Account — initiative_type Enum DIGITAL_ONBOARDING / SME_LENDING / NEO_EXPANSION INIT-RECORD-0 / OPEN_BANKING / POS_FINANCE / OTHER 01 maturity_stage Enum ANNOUNCED / PLANNING / BUILDING / LIVE / SCALING INIT-MATURITY- 001

MARKETMEM Rules (5)

MARKETMEM-RECORD-001

Market_Interaction_History aggregates all interactions across accounts and time, keyed to persistent Contact identity, forming Decimal's market reputation memory.

Rationale: *#2 — Why track interactions across career transitions? MARKETMEM-002 requires knowing if Decimal engaged this PERSON before, anywhere. Career moves are where relationship capital transfers; losing this history loses the relationship.*

MARKETMEM-CAREER-001

When a Contact changes employer, their full interaction history is preserved and surfaced in the new account context. Prior rapport (or damage) carries forward.

Rationale: *#3 — Why compute a reputation blast-radius per contact? High-influence contacts shape how a whole segment perceives Decimal. A mistake with them propagates. Blast-radius sizing tells the system where maximum care is required.*

MARKETMEM-BLAST-001

Each interaction estimates a reputation blast_radius based on contact influence and network position. High-blast contacts mandate maximum-tier handling and conservative outreach.

Rationale: *#4 — Why track brand-capital depletion signals here? Repeated negative interactions are an early warning of reputation damage. Aggregating them market-wide surfaces depletion before it becomes a crisis (the Brand Capital Depletion Alert).*

MARKETMEM-DEPLETION-001
Market_Interaction_History monitors depletion signals (rising negative sentiment, ignored outreach, complaints). Two-or-more triggers pause market expansion pending review and repair.
Rationale: *#5 — Deepest reason this entity exists? It is the system's conscience and long memory — the reason it behaves like a trusted market participant rather than a spam engine. It exists because reputation, once spent, is the hardest capital to rebuild.*

MARKETMEM-PRIMACY-001
Every outreach decision consults Market_Interaction_History for the target person. Prior unresolved damage is a hard gate; the system never re-burns a known-damaged relationship. MARKET_INTERACTION_HISTORY — Field Specification Field Type Business Rule Source history_id UUID Immutable ledger entry MARKETMEM-R ECORD-001 contact_id UUID Persistent person key MARKETMEM-C AREER-001 account_context_id UUID Account at time of interaction MARKETMEM-R ECORD-001

MESSAGE Rules (3)

MESSAGE-RECORD-001
Every outbound message — sent, queued, or QC-rejected — is stored with full content, channel, sequence position, and the Reasoning_Chain that produced it. Rejected drafts retained for failure analysis.
Rationale: *#2 — Why link every Message to its Reasoning_Chain? Principle 5 forbids generating messages from raw data — every message must have a traceable thought process. Without the link you can't diagnose whether a bad outcome came from bad reasoning or bad execution.*

MESSAGE-TRACE-001
No Message may exist without a reasoning_chain_id. A message with no reasoning lineage is rejected before send.
Rationale: *#3 — Why store prompt_version and AI metadata on Message? Messages are generated by evolving prompts. To optimize and enable rollback, each message records which prompt version and model produced it — tying outcomes to specific generation logic.*

MESSAGE-OUTCOME-001
Each Message records response_received, response_sentiment, response_latency. Feeds Outcome_Attribution and Feedback_Loop_Entry; sentiment also updates Contact.interaction_lineage.
Rationale: *#5 — Deepest reason Message exists? Message is where strategy meets reality and the system is most accountable — the point of brand-capital expenditure. Storing it fully keeps the system honest and prevents repeating a mistake on the same person.*

MODELREG Rules (4)

MODELREG-RECORD-001
Each AI model and version is registered with capabilities, intended decision types, and activation status. Generated artifacts record the model_id that produced them.
Rationale: *#2 — Why track per-model performance? A model upgrade can silently degrade quality. Per-model performance metrics catch regressions and inform which model handles which task.*

MODELREG-PERF-001
Performance metrics (quality, hallucination rate, outcome correlation) are tracked per model/version. Regression beyond thresholds blocks promotion and can trigger rollback.
Rationale: *#3 — Why route decision types to specific models? Different tasks suit different models (cheap/fast for parsing, strong for strategy). Explicit routing optimizes cost and quality.*

MODELREG-ROUTING-001
Decision types route to designated models per the registry. Routing changes are versioned and auditable; a model may only handle decision types it is approved for.
Rationale: *#4 — Why require rollback capability? If a new model misbehaves in production, the system must revert instantly. Rollback is a safety requirement, not a nicety.*

MODELREG-ROLLBACK-001
Every model activation is reversible. A one-action rollback restores the prior approved model for affected decision types; rollback events are logged as System_Events.

PATTERN Rules (4)

PATTERN-RECORD-001
Recurring decision contexts and their outcomes are abstracted into Decision_Pattern entities (situation signature, decision taken, outcome distribution, confidence).
Rationale: *#2 — Why require an outcome distribution, not a single result? One success can be luck. A pattern is trustworthy only across many instances. Storing the distribution captures reliability, not anecdote.*

PATTERN-EVIDENCE-001
Each pattern carries the count and outcome distribution of instances. Patterns below a minimum instance threshold are advisory only and cannot auto-drive decisions.
Rationale: *#3 — Why link patterns to the rules they support or challenge? Patterns are how rules are validated empirically. A pattern that contradicts a rule is a signal the rule needs revisiting; one that confirms it strengthens confidence.*

PATTERN-DECAY-001
Pattern confidence decays without fresh confirming instances. Stale patterns are demoted to advisory and re-validated before reuse in automated decisions.
Rationale: *#5 — Deepest reason this entity exists? It is how the system gets wiser over time rather than merely busier — the accumulation of strategic experience. It exists to turn outcomes into reusable intelligence, the essence of a learning system.*

PATTERN-PRIMACY-001
Decision patterns inform but never silently override explicit rules or human judgment. Auto-application requires sufficient evidence and current confidence; otherwise patterns are presented as recommendations. DECISION_PATTERN — Field Specification

PROMPT Rules (6)

PROMPT-RECORD-001
Each prompt template version is stored with its text, target decision type, and model compatibility. Generated artifacts record prompt_version for traceability.
Rationale: *#2 — Why tie outcomes back to prompt versions? To know which phrasing converts, outcomes must attribute to the prompt that produced them. Without this link, prompt optimization is guesswork.*

PROMPT-OUTCOME-001
Outcomes (response rate, sentiment, conversion) attribute to the prompt_version via Message and Outcome_Attribution, enabling evidence-based prompt optimization.
Rationale: *#3 — Why support A/B testing of prompts? Improvement requires controlled comparison. The entity must allow concurrent versions with traffic splits and significance tracking.*

PROMPT-AB-001
Prompt versions support controlled A/B allocation with sample-size and significance tracking. Winning variants promote only on statistically meaningful improvement, not noise.
Rationale: *#4 — Why require rollback and approval for prompt changes? A bad prompt can damage brand at scale instantly. Changes need approval and instant revert, like any production change touching customers.*

PROMPT-GOVERNANCE-001
Prompt promotion to production requires approval and is reversible. Prompts affecting maximum-tier contacts undergo human review before activation.
Rationale: *#5 — Deepest reason this entity exists? It treats the system's words — its actual voice to the market — as a governed, improvable, accountable asset. It exists because in this system, language is the product surface, and it must be controlled.*

PROMPT-PRIMACY-001
No Message may be generated by an unregistered prompt. Every AI-generated artifact links to a valid Prompt_Version; untracked prompt use is prohibited.

PROMPT-AB-00
1 active / approved_by Bool/UUID Governance status PROMPT-GOVE RNANCE-001 ENTITY 29: CAPABILITY_GAP A recorded instance where the system recognized it lacked the data, skill, or authority to act well — institutionalized humility.
Rationale: *#1 — Why record what the system CANNOT do as an entity? A system that hides its blind spot*

REVIEW Rules (2)

REVIEW-RECORD-001
Each human review is stored with the artifact reviewed, the decision (approve/edit/reject), edits made, rationale, and reviewer identity. Reviews are immutable once submitted.
Rationale: *#2 — Why capture the rationale, not just the verdict? A bare reject teaches the system nothing. The rationale is the training signal that improves future AI proposals and detects rubber-stamping.*

REVIEW-RATIONALE-001
Approvals and rejections require a rationale. Rationale-free verdicts are flagged; high rates of rationale-free approvals signal rubber-stamping and trigger oversight review.
Rationale: *#3 — Why monitor for rubber-stamping explicitly? Rubber-stamping is a named failure mode that hollows out oversight. The system must detect reviewers who approve everything instantly without engagement.*

SCORE Rules (1)

SCORE-RECORD-001
Each scoring run writes an immutable Three_Score_Record snapshot (fit/ICS, intent/dynamic, intelligence-confidence) with per-dimension contributions and the weight-set version used. The latest snapshot updates Account.current_state.

SEQ Rules (5)

SEQ-RECORD-001
Each Outreach_Sequence stores ordered steps, channel per step, timing gaps, and stop conditions. It enforces hard limits: max touches per channel, minimum gaps, and the 14/10-day spreads.
Rationale: *#2 — Why encode stop/exit conditions in the sequence? A sequence that ignores responses is spam. It must halt on reply, on negative sentiment, or on window closure — exit logic is core, not optional.*

SEQ-EXIT-001
Each sequence defines exit conditions (reply received, negative sentiment, window closed, do-not-contact). Triggering any exit immediately pauses/terminates remaining steps.
Rationale: *#3 — Why bind sequence pacing to system-wide rate limits? Per-contact limits aren't enough; the system has global throughput caps (max new accounts/day in MVP). The sequence must respect both contact-level and system-level pacing.*

SEQ-RATELIMIT-001
Sequence scheduling respects contact-level limits AND global throughput caps. When global capacity is exhausted, sequence starts queue by priority (Decision Score), not first-come.
Rationale: *#4 — Why make each step re-evaluate before firing? State changes between planning and execution (window closes, contact leaves, sentiment drops). A step must re-check conditions at fire-time, not blindly execute the plan.*

SEQ-REEVAL-001
Before each step fires, the system re-evaluates current account/contact/window state and the governing Reasoning_Chain freshness. Stale or invalidated steps are regenerated or cancelled.
Rationale: *#5 — Deepest reason this entity exists? It operationalizes timing and restraint — turning strategy into paced, condition-aware action instead of a burst of messages. It exists because in enterprise, cadence and patience are as important as content.*

SEQ-PRIMACY-001
All outbound activity flows through an Outreach_Sequence governed by these constraints. Ad-hoc out-of-sequence sends are prohibited except explicit human-initiated one-offs, which are still logged. OUTREACH_SEQUENCE — Field Specification

SIGNAL Rules (4)

SIGNAL-PERSISTENCE-001
Every signal is stored immutably with raw source payload, detection timestamp, and source attribution — even after scoring/action. Signals decay in weight, not existence; never deleted.
Rationale: *#2 — Why store the raw payload, not just the parsed signal? Parsing logic evolves. A signal parsed under v1 may need reinterpretation under v2. Raw payload allows reprocessing and protects against parser bugs corrupting evidence.*

SIGNAL-PROVENANCE-001
Signal stores raw_payload (immutable) and parsed_interpretation (versioned, parser_version). Reparsing creates a new interpretation linked to the same raw payload; original retained.
Rationale: *#3 — Why a decay model on the entity itself? Timing is the primary competitive weapon. A CTO-hiring signal is P1-hot at week 1 and noise at month 6. Current weight is a function of age, so the decay curve is a property of signal type, computed at read time.*

SIGNAL-DEDUP-001
Each Signal computes an event_fingerprint (entity + event_type + time-window). Signals sharing a fingerprint collapse into one logical signal with multiple corroborating_sources; corroboration raises confidence but does NOT multiply weight.

SIGNAL-PRIMACY-001
No account may enter or advance in outreach without at least one qualifying, non-expired, deduplicated signal. The triggering signal is permanently linked to the resulting Reasoning_Chain. SIGNAL — Field Specification Field Type Business Rule Source signal_id UUID Immutable SIGNAL-PERSIS TENCE-001 account_id UUID FK to Account (or Unresolved_Intelligence) ACCT-IDENTITY -005 signal_type Enum Drives priority + decay curve SIGNAL-DECAY -001 priority Enum P1 / P2 / P3 base weight SIGNAL-DECAY -001 raw_payload Object Immutable source data SIGNAL-PROVE NANCE-001 parsed_interpretation Object Versioned parse (parser_version) SIGNAL-PROVE NANCE-001 event_fingerprint String Dedup key SIGNAL-DEDUP -001 corroborating_sources Array Multi-source confirma...

VENDOR Rules (3)

VENDOR-SATISFACTION-001
Dissatisfaction signals (complaints, failed projects, public RFPs) are tracked against incumbents. Dissatisfaction + approaching renewal opens a displacement Buying_Window.
Rationale: *#4 — Why track contract/renewal timing? Displacement is only realistic near renewal. Knowing renewal timing converts a hopeless 'rip and replace' into a well-timed competitive entry.*

VENDOR-RENEWAL-001
Known contract/renewal dates are stored and feed window prediction. Outreach ramps as renewal approaches; mid-contract displacement attempts are de-prioritized absent strong dissatisfaction.
Rationale: *#5 — Deepest reason this entity exists? It is the system's competitive awareness — the difference between selling into reality and selling into a fantasy of empty space. It exists because every enterprise account already has incumbents.*

VENDOR-PRIMACY-001
Account strategy must state the incumbency posture (greenfield / coexist / displace / wait) backed by Vendor_Intelligence. An unstated competitive posture lowers strategy confidence. VENDOR_INTELLIGENCE — Field Specification Field Type Business Rule Source vendor_intel_id UUID Immutable VENDOR-RECO RD-001 account_id UUID FK to Account — incumbent_name / String/Object What they run today VENDOR-RECO product_footprint RD-001

Section 3: Modules and Workflows (61 rules)

M0 Rules (5)

M0-GATE-001
No intelligence (signal, enrichment, engagement) may be consumed by a downstream module until M0 has assigned it an Intelligence Confidence Score and passed it through freshness, dedup, and provenance checks. Unblessed data is quarantined, never silently used.
Rationale: *#2 — Why compute confidence as a SCORE rather than a pass/fail flag? Because data quality is a gradient, not a binary. A two-week-old firmographic fact is usable but weaker than a fresh one. A score lets downstream modules weight intelligence proportionally (ICS gates intent in the opportunity equation) instead of crudely accepting or rejecting it.*

M0-CONF-001
M0 computes a three-dimensional Intelligence Confidence Score (source reliability x freshness x corroboration) per intelligence item. The score flows into Three_Score_Record as the confidence dimension and proportionally weights the item everywhere it is used.
Rationale: *#3 — Why route conflicts and gaps through M0 rather than each module handling its own? Because consistency requires one authority. If every module resolves conflicting enrichment its own way, the system contradicts itself. M0 centralizes conflict resolution and gap declaration so the whole system reasons from one reconciled truth and one honest list of what it does not know.*

M0-RECONCILE-001
M0 owns intelligence-conflict reconciliation (via ENRICH-RECONCILE-001) and Capability_Gap declaration. When sources disagree or data is missing, M0 records the conflict/gap and emits a confidence-lowered or quarantined result rather than letting modules guess independently.
Rationale: *#4 — Why must M0 be event-driven rather than a periodic batch cleaner? Because timing is the primary weapon — a hot signal blessed six hours late is a missed window. M0 must evaluate intelligence the moment it arrives so downstream reaction is immediate, while still catching the staleness of older items at read time.*

M0-EVENT-001
M0 evaluates each intelligence item on arrival (event-driven), emitting a System_Event on bless/quarantine. It also re-evaluates confidence at read time so decay since ingestion is always reflected. No fixed-interval batch gate is permitted for hot signals.
Rationale: *#5 — Deepest reason M0 exists? Because the system's single greatest risk is acting confidently on bad information — the hallucination/staleness/metric-manipulation failure family. M0 is the institutional discipline that the system never mistakes noise for knowledge. It is the foundation of trust for everything above it.*

M0-PRIMACY-001
M0 is the first module in the pipeline and an immutable governance authority. Its confidence scores and quarantine decisions cannot be overridden by downstream modules; only a governed human action can release quarantined intelligence. ① Workflow Logic (step-by-step) 1. Intelligence item arrives (signal / enrichment / engagement) → emit ingest System_Event 2. Provenance check: is raw_payload + source attribution present? (SIGNAL-PROVENANCE / ENRICH-RECORD) 3. Dedup check: compute event_fingerprint; collapse to corroborating_sources if matched 4. Freshness check: apply decay/staleness window for the item type

M1 Rules (3)

M1-NORMALIZE-001
M1 normalizes every raw source payload into the Signal schema (signal_type, priority, raw_payload, parsed_interpretation, fingerprint) with parser_version stamped. Raw payload is always retained for reparsing under future parser versions.

M1-RESOLVE-001
M1 performs entity resolution against Account.aliases. Resolved signals attach to the Account; unresolved signals go to Unresolved_Intelligence for later resolution (per ACCT-IDENTITY-005). Misattribution is worse than non-attribution — ambiguous matches stay unresolved pending confidence.
Rationale: *#4 — Why does M1 hand off to M0 rather than directly trigger scoring? Because raw detection is not blessed intelligence. M1 finds and structures; M0 judges quality. Separating detection from governance prevents a noisy or duplicate signal from triggering a scoring cascade before its confidence is established.*

M1-HANDOFF-001
M1 emits each normalized signal to M0 for governance, not directly to M2 scoring. Scoring is triggered only after M0 blesses the signal. M1 never assigns confidence or triggers downstream action itself.
Rationale: *#5 — Deepest reason M1 exists? Because the system can only be as aware as its senses. M1 is the sensory organ — without continuous, structured, attributed perception of the market, every other module is reasoning in the dark. It is the physical embodiment of event-driven architecture.*

M10 Rules (1)

M10-RECORD-001
M10 records every human review as an immutable Human_Review_Record (artifact, decision approve/edit/reject, edits, rationale, reviewer, latency). It routes qualified engagement to the right human role (SDR/AE/technical) by trigger type (REVIEW-RECORD-001).
Rationale: *#2 — Why require rationale on every review decision, not just the verdict? Because a bare approve/reject teaches the system nothing and hides rubber-stamping. Rationale is the training signal that improves future AI proposals and lets the system detect oversight that has gone hollow.*

M11 Rules (3)

M11-CENTRAL-001
M11 is the sole authority for system self-adjustment. Every change (scoring weight, threshold, timing curve, prompt promotion, pattern update) is a Feedback_Loop_Entry citing evidence, rate-limited and reversible (FEEDBACK-RECORD-001, FEEDBACK-PRIMACY-001).
Rationale: *#2 — Why attribute outcomes multi-touch rather than last-touch? Because enterprise wins come from many touches over time; last-touch attribution misleads and would teach the system the wrong lessons. Multi-touch credit reveals which sequence of decisions actually built the outcome.*

M11-ATTRIBUTE-001
M11 produces Outcome_Attribution with multi-touch credit across contributing Reasoning_Chains, Prompt_Versions, timing, and connector paths (ATTR-MULTITOUCH-001). Failures are attributed with equal rigor (ATTR-FAILURE-001).

M11-BASELINE-001
M11 maintains Performance_Baselines per metric and segment, continuously compares live metrics, and on significant regression alerts and can auto-pause/roll-back the responsible change (BASELINE-REGRESSION-001). Re-baselining is an explicit logged action (BASELINE-REBASE-001).
Rationale: *#5 — Deepest reason M11 exists? Because the system's deepest purpose is to accumulate judgment, not just activity. M11 is the disciplined heartbeat of learning — evidence-based, bounded, auditable, reversible improvement that makes the system measurably wiser over time without betraying its principles.*

M12 Rules (1)

M12-DNA-001
M12 maintains Organizational_DNA as a slow prior, updating only when Organizational_Assessment deviations persist across multiple readings (DNA-PRIOR-001). It clusters DNA into reusable buying archetypes and assigns provisional archetypes to sparse accounts (DNA-ARCHETYPE-001).
Rationale: *#3 — Why does M12 own market-level narrative tracking? Because narrative is the only supra-account entity — it shapes the whole segment, not one account. A dedicated owner tracks narrative lifecycle and computes per-account alignment, feeding framing and timing across all accounts simultaneously.*

M13 Rules (2)

M13-LTV-001
M13 estimates expected lifetime value including expansion potential and reference value (linked to Market_Dependency_Graph). Opportunities that are keystones or platform-entry points carry LTV above their initial deal size; this feeds M6 keystone valuation.
Rationale: *#5 — Deepest reason M13 exists? Because a system that allocates finite capital without modeling economic value is flying blind on the one dimension that ultimately matters — return. M13 gives the system economic gravity: the sense that not all opportunities deserve equal capital because not all are worth the same. It turns prioritization into investment.*

M13-PRIMACY-001
No opportunity may be allocated significant capital without an M13 economic profile (or an explicit unknown-value flag that caps the capital committed). Value estimates carry confidence; low-confidence estimates cap committed capital until refined. ① Workflow Logic (step-by-step) 1. Opportunity created/updated by M19 → request economic profile 2. Estimate deal size from opportunity type, account size, initiative criticality, comparable deals 3. Estimate budget probability from funding signals, sponsorship, org stability 4. Estimate realization probability + expected LTV (expansion + reference value) 5. Compute expected economic value (size x budget-prob x realization) with confidence 6. Emit value term to M2 (opportunity equation) and M6 (E...

M14 Rules (5)

M14-DYNAMICS-001
M14 maintains competitive dynamics per Opportunity: competitor_strength, displacement_probability, incumbent_satisfaction, replacement_triggers, political_protection_index. These are live, evolving assessments distinct from M3's static Vendor_Intelligence facts.
Rationale: *#2 — Why model political protection specifically? Because in KSA banking, an incumbent can be technically weak yet politically immovable (a relationship-based or regulator-favored vendor), or technically strong yet politically exposed. Ignoring political protection makes the system pursue technically-winnable but politically-impossible displacements — burning capital on dead ends.*

M14-POLITICAL-001
M14 maintains a political_protection_index per incumbent (relationship depth, regulatory favor, executive sponsorship of the incumbent). High protection suppresses displacement_probability even when technical vulnerability is high; the system avoids politically-impossible displacements.
Rationale: *#3 — Why track replacement triggers rather than just current state? Because displacement happens at moments — a failed project, a renewal, a leadership change, a public RFP. Knowing the triggers lets the system wait for the opening rather than pushing against a stable incumbent. This is timing-as-weapon applied to competition.*

M14-TRIGGER-001
M14 tracks replacement triggers (incumbent failure, renewal approach, sponsor departure, public RFP, dissatisfaction signals). A trigger firing opens or amplifies a displacement Buying_Window (via M5) and raises displacement_probability. Absent triggers, entrenched incumbents are de-prioritized.
Rationale: *#4 — Why feed competitive dynamics into strategy (M4) and simulation (M15)? Because the right strategy depends on the incumbent's likely response. A vulnerable incumbent invites direct displacement; a protected one demands coexistence or patience. M15 stress-tests strategy against incumbent retaliation — which requires M14's model of how the incumbent will react.*

M14-STRATEGY-001
M14 dynamics condition M4 competitive posture (displace/coexist/wait/greenfield) and feed M15 simulation as the incumbent-response model. A strategy assuming an incumbent will not react, against a strong/protected incumbent, is flagged as unrobust.
Rationale: *#5 — Deepest reason M14 exists? Because enterprise selling is usually a contest against an incumbent, and a system that cannot read the battlefield will pick the wrong fights. M14 gives the system competitive awareness — the difference between selling into a fantasy of empty space and selling into a contested, dynamic, political reality.*

M14-PRIMACY-001
Any displacement strategy must be backed by an M14 competitive assessment. Pursuing displacement without modeling incumbent vulnerability and political protection is prohibited; the default assumption is that every enterprise opportunity is contested. ① Workflow Logic (step-by-step) 1. Opportunity with incumbent identified (from M19/M3) 2. Assess competitor_strength + incumbent_satisfaction from signals/enrichment 3. Compute political_protection_index (relationships, regulatory favor, sponsorship)

M15 Rules (2)

M15-ROBUST-001
M15 selects the strategy that is robust across scenarios (strong expected outcome with bounded downside), not merely the highest best-case outcome. A high-mean / catastrophic-downside strategy is rejected in favor of a robust alternative. Downside weighting reflects brand/relationship capital at risk.

M15-CALIBRATE-001
M15 outputs are ADVISORY (labeled estimates) until M11 calibrates scenario predictions against observed outcomes. Until calibrated, M15 may recommend and flag fragile strategies but may not auto-reject an M4 strategy without human review. Calibration lifts this restriction per decision type.
Rationale: *#5 — Deepest reason M15 exists? Because mature strategy is not choosing the best plan — it is choosing the plan that survives the buyer reacting. M15 moves the system from picking a strategy to pressure-testing it, from optimism to robustness. It is the difference between a planner and a strategist.*

M16 Rules (3)

M16-TRACK-001
M16 processes conversational replies (post-M9) to extract structured signals: objections, commitments, sentiment shifts, and newly-emerged stakeholders. These enrich the Opportunity's committee and feed M10 handoff with conversation context, not just a 'positive reply' flag.
Rationale: *#2 — Why track stakeholder EMERGENCE during conversation? Because buying committees reveal themselves through dialogue — a CTO loops in security, procurement appears, a new sponsor is cc'd. These emergent stakeholders must be added to the opportunity's committee in real time, or the committee map goes stale exactly when the deal heats up.*

M16-STAKEHOLDER-001
Stakeholders emerging in conversation are added to the Opportunity's Buying_Committee (via M19/M3) with their observed stance. The committee map updates live during active conversations; emergence is a first-class signal that can reshape strategy.
Rationale: *#3 — Why must M16 be scoped to TRACKING before AUTONOMOUS handling at MVP? Because the FSD explicitly excludes autonomous negotiation from MVP, and autonomously conducting enterprise conversations carries high brand/relationship risk before the system has earned that trust. M16 starts by extracting and surfacing conversation intelligence FOR the human; autonomous handling is a later, trust-gated expansion.*

M16-LEARN-001
M16 emits objection/commitment/resolution data to M11 (attribution + pattern learning) and M17 (expert heuristics). Recurring objections inform Decision_Patterns and message strategy; unresolved objection types surface as Capability_Gaps.

M17 Rules (4)

M17-EXTRACT-001
M17 analyzes Human_Review_Records to detect recurring reviewer patterns and abstracts them into named heuristics (e.g. 'Reviewer-X Heuristic #14: reject engagement when readiness < 60'). Heuristics carry the evidence count and the reviewer they derive from.
Rationale: *#2 — Why attribute heuristics to specific reviewers rather than blend them? Because reviewers have different expertise and sometimes disagree. Blending them into one anonymous policy loses the signal of WHO is reliable on WHAT, and hides genuine expert disagreement that itself is informative. Attributed heuristics preserve the source of judgment.*

M17-ATTRIBUTE-001
Heuristics are attributed to their source reviewer and decision domain. Conflicting heuristics between reviewers are retained and surfaced (not silently merged), preserving expert disagreement as signal. Reviewer reliability per domain (from M10 outcomes) weights heuristic confidence.
Rationale: *#3 — Why promote stable heuristics toward Decision_Patterns and candidate rules? Because a heuristic confirmed across many decisions and validated by outcomes is institutional knowledge that should inform the system broadly — not stay locked to one reviewer's queue. Promotion is how individual expertise becomes organizational capability.*

M17-FEEDBACK-001
Validated heuristics inform M4 reasoning (so AI proposals pre-satisfy known reviewer expectations) and M10 review routing (auto-handling cases that established heuristics clearly resolve, escalating genuinely novel ones). This reduces rubber-stamping pressure by removing trivial reviews.
Rationale: *#5 — Deepest reason M17 exists? Because an organization's hardest-won asset is the judgment of its experts, and a system that lets that judgment die in individual decisions wastes it. M17 institutionalizes expertise — turning what a reviewer knows into what the system knows. It is the human-side mirror of strategic memory.*

M17-PRIMACY-001
M17 converts human judgment into reusable institutional knowledge under governance. Extracted heuristics inform but never silently override explicit rules or current human review; promotion to rules is always governed (parallels PATTERN-PRIMACY-001). ① Workflow Logic (step-by-step) 1. Human_Review_Records accumulate from M10 2. Mine rationales for recurring decision patterns per reviewer + domain 3. Abstract stable patterns into named, attributed heuristics with evidence count 4. Validate heuristics against eventual outcomes (reviewer reliability weighting) 5. Surface conflicting heuristics between reviewers (do not merge) 6. Promote outcome-validated heuristics → Decision_Patterns / candidate rules (governed) 7. Feed validated heuristics t...

M18 Rules (2)

M18-OBSERVE-001
M18 ingests and learns from third-party market events (competitor wins, public procurement awards, industry announcements, implementation successes/failures) that do not involve Decimal. These become learning signal alongside Decimal's own outcomes (M11).
Rationale: *#2 — Why does market learning especially strengthen Tier 6 (M12)? Because narrative, dependency, and DNA are market-level judgments best learned from market-wide events. A competitor's win at one bank predicts narrative spread and reference effects across the segment; an implementation failure teaches which approaches fail. M18 is the primary external feed into strategic memory.*

M18-COMPETITIVE-001
M18 routes observed competitor wins/losses/failures to M14, updating competitor strength, displacement probability, and replacement triggers from real market outcomes rather than only account-local signals.
Rationale: *#5 — Deepest reason M18 exists? Because a system that learns only from its own actions learns slowly and narrowly, while the market runs a continuous experiment it could learn from for free. M18 turns the whole market into a teacher — the single biggest accelerant to the system's judgment. It is learning at market scale.*

M19 Rules (4)

M19-GRAPH-001
M19 owns construction and maintenance of the per-account opportunity graph: it creates Opportunity entities, assigns committees/windows/incumbents/budgets to the correct Opportunity, and maintains influence paths between nodes. M3 supplies facts; M19 partitions them into journeys.
Rationale: *#2 — Why model stakeholders, vendors, and budgets as NODES rather than fields? Because the same stakeholder can influence multiple opportunities, the same incumbent can defend several, and budgets can be shared or contested across journeys. A node-and-edge graph captures these many-to-many realities that flat fields on a single Opportunity cannot.*

M19-NODES-001
The opportunity graph contains typed nodes (Opportunity, Stakeholder, Vendor, Budget, Initiative) and influence edges. A stakeholder node may connect to several Opportunity nodes; a budget node may fund or be contested across journeys. Cross-opportunity influence is first-class.

M19-RESOLVE-001
M19 resolves each blessed signal to the correct Opportunity within the account (creating a new Opportunity when none fits). Ambiguous signals attach at account level pending resolution, never mis-assigned to an arbitrary journey. This is opportunity-resolution, parallel to M1 entity-resolution.
Rationale: *#4 — Why does M19 sit upstream of scoring (M2) and strategy (M4) now? Because M2 and M4 now operate per opportunity — they cannot score or strategize until M19 has established WHICH opportunity the intelligence belongs to. M19 becomes part of the mandatory pipeline: detect → bless → resolve-to-opportunity → score → reason.*

M19-PRIMACY-001
M19 is the authority on opportunity structure within an account. Account-level roll-ups are derived from M19's graph; no module may treat an account as a single opportunity when M19 has decomposed it into several. ① Workflow Logic (step-by-step) 1. Blessed signal / enrichment arrives for an account (from M0/M3) 2. Opportunity-resolution: does this belong to an existing Opportunity, or spawn a new one? 3. If new → create Opportunity (type, linked initiatives); if existing → attach 4. Assign/Update nodes: stakeholders → committees, vendors → incumbents, budgets → funding 5. Maintain influence edges (cross-opportunity stakeholder/budget links) 6. Emit per-Opportunity events to M2 (score), M5 (window), M4 (strategy), M13 (economics) 7. Maintain...

M2 Rules (1)

M2-PRIMACY-001
No sequencing, allocation, or outreach may reference a priority not produced and stored by M2. Rankings on ephemeral or module-local scores are prohibited (per SCORE-PRIMACY-001). ① Workflow Logic (step-by-step) 1. Subscribed event received (blessed signal / window change / engagement / readiness change)

M3 Rules (3)

M3-TRIGGER-001
M3 enriches on M2 HOT-tier detection (or explicit human request). Speculative enrichment of COLD accounts is rate-limited and requires justification. Enrichment scope scales with Decision Score — higher score earns deeper enrichment.

M3-GOVERN-001
M3 submits enriched facts to M0 for provenance, freshness, and conflict reconciliation before they are usable. Each fact carries source + source_date; conflicts are reconciled by source reliability with both values retained (ENRICH-RECONCILE-001).
Rationale: *#4 — Why does M3 feed both scoring (M2) and strategy (M4)? Because enrichment changes both how attractive an account is and how to approach it. New committee data and initiative intelligence retrigger scoring AND inform the reasoning chain. M3 is the bridge that deepens both prioritization and strategy.*

M3-FEED-001
On enrichment completion, M3 emits events that retrigger M2 scoring (new dimensions available) and notify M4 (committee + initiative + vendor intelligence ready for strategy). Enrichment is not complete until these events fire.
Rationale: *#5 — Deepest reason M3 exists? Because the system must convert the unknown into the known before it can act wisely. M3 is the research faculty — the difference between reaching out to a name and reaching out to a understood organization with a mapped committee and a known initiative. It earns the right to personalize.*

M4 Rules (5)

M4-SEPARATE-001
M4 produces the Reasoning_Chain and Account_Strategy BEFORE M7 generates any message. Generation without a complete, M4-produced reasoning chain is rejected (per REASON-PRIMACY-001 and MESSAGE-TRACE-001).
Rationale: *#2 — Why must M4 maintain a standing Account_Strategy above individual reasoning chains? Because sequences are tactics and the account needs one coherent strategy they all serve. Without a standing strategy, two touches can contradict each other and the account experiences an incoherent Decimal. The strategy holds the angle, posture, committee targets, connector path, and readiness gates.*

M4-STRATEGY-001
M4 maintains one active Account_Strategy per engaged account (value angle, competitive posture, target committee members, connector path, readiness gates, anchor initiative/window). All Reasoning_Chains and downstream sequences must conform to it (per STRAT-PRIMACY-001).
Rationale: *#3 — Why does M4 consume Organizational_DNA and Market_Narrative, not just signals? Because strategy is about character and context, not just events. The same signal demands a different strategy for a slow-consensus, governance-heavy account (SNB pattern) than for a fast-mover, and a different framing inside an accelerating narrative than a vacuum. M4 conditions its reasoning on Tier 6 judgment.*

M4-CONDITION-001
M4 conditions strategy on Organizational_DNA (pacing, consensus difficulty, displacement feasibility) and Market_Narrative alignment/phase (framing, urgency). A strategy contradicting known DNA (e.g. fast-close on a slow-consensus account) is flagged for justification (DNA-PRIMACY-001).

M4-CHAIN-001
M4 executes the mandatory reasoning sequence (signal interpretation → account research → pain inference → persona analysis → strategy decision → [handoff to M7 generation] → QC) storing each step's input/output/confidence. Any skipped or low-confidence step invalidates the chain (REASON-STEPS-001).
Rationale: *#5 — Deepest reason M4 exists? Because the difference between an intelligent system and a template engine is whether there is a defensible thought process behind every action. M4 is the strategic mind — it ensures the system always knows WHY it is about to act, can explain it, and can improve it. It is reasoning made institutional.*

M4-PRIMACY-001
No message, sequence step, or strategic action may exist without an M4-produced, complete, QC-passing Reasoning_Chain conforming to the active Account_Strategy. This is enforced at the data layer, not by convention. ① Workflow Logic (step-by-step) 1. Enrichment-ready event from M3 (or strategy-review trigger) received 2. Step 1 Signal interpretation: what happened, why it matters to Decimal 3. Step 2 Account research: priorities, initiative, incumbency, DNA, narrative alignment 4. Step 3 Pain inference: operational pains implied by signals + research 5. Step 4 Persona analysis: target committee member motivations, decision style, risk posture 6. Step 5 Strategy decision: angle, value prop, ask, connector path, timing intent 7. Build/refresh...

M5 Rules (3)

M5-PREDICT-001
M5 maintains estimated_closure with confidence per window, updated as signals age and via M11 timing calibration. Approaching closure raises execution urgency in the decision sequence and can pre-empt lower-priority work (WINDOW-PREDICT-001).
Rationale: *#3 — Why does M5 read Market_Narrative phase, not just account signals? Because a window opened by a signal riding an accelerating narrative is stronger than the same signal in a vacuum. Narrative phase amplifies or dampens the window — timing is about the market's tempo, not just the account's.*

M5-GATE-001
M5 provides the timing decision in the Universal Decision Sequence. No 'now vs later' decision may be made without consulting the account's active window state (WINDOW-PRIMACY-001). Closing windows can re-prioritize execution order ahead of higher raw scores.
Rationale: *#5 — Deepest reason M5 exists? Because in enterprise selling, being right at the wrong time is being wrong. M5 is the system's sense of timing — the discipline that turns 'who to contact' into 'who to contact NOW.' It is timing-as-weapon made into running code.*

M5-PRIMACY-001
M5 is the authority on engagement timing. Execution (M8) may not start or advance a sequence step against M5's window/timing guidance without explicit override. Timing is enforced, not advisory. ① Workflow Logic (step-by-step)

M7 Rules (5)

M7-SEPARATE-001
M7 generates message content strictly FROM an M4 Reasoning_Chain; it may not introduce strategy or facts absent from the chain. Each generated Message links its reasoning_chain_id, prompt_version, and model_id (MESSAGE-AI-PROVENANCE-001).
Rationale: *#2 — Why must every generation pass QC before it can be queued? Because hallucination is a named failure mode and a single fabricated fact can destroy credibility with an auditor-minded buyer. QC (hallucination score, factual grounding, tone, enterprise suitability) is a hard gate, not a suggestion.*

M7-QC-001
M7 runs mandatory QC on every draft: hallucination_score, factual-grounding against blessed enrichment, tone, and enterprise suitability. Drafts failing thresholds are rejected (status QC_REJECTED, retained for analysis) and never queued (REASON-QC-001).
Rationale: *#3 — Why ground every claim against M0-blessed enrichment specifically? Because a message may only assert what the system actually, verifiably knows. Grounding generation against blessed facts (not the model's parametric guesses) is how the system stays truthful with high-stakes buyers.*

M7-GROUND-001
Every factual claim in a generated message must trace to an M0-blessed Enrichment fact or Signal. Ungrounded claims are stripped or the draft is rejected. The model may not introduce unverifiable specifics (numbers, names, dates) absent from the intelligence base.
Rationale: *#4 — Why version prompts and route generation through the registry? Because phrasing performance must be measurable and reversible. Tying each message to a Prompt_Version enables A/B testing, outcome attribution, and instant rollback of a prompt that damages brand at scale.*

M7-PROMPT-001
M7 generates only via registered Prompt_Versions and approved AI_Model_Registry models for the decision type. Outcomes attribute to prompt_version for M11 optimization; prompts affecting max-tier contacts require human approval before activation (PROMPT-GOVERNANCE-001).

M7-PRIMACY-001
No Message exists without an M7 generation linked to a Reasoning_Chain and a registered Prompt_Version, having passed QC. Untracked or un-QC'd message creation is prohibited (MESSAGE-TRACE-001, PROMPT-PRIMACY-001). ① Workflow Logic (step-by-step) 1. Generation request from M4 with completed Reasoning_Chain + target channel/persona/language 2. Select registered Prompt_Version + approved model for the decision type 3. Generate draft content grounded strictly in chain + blessed enrichment 4. QC: hallucination score, factual grounding, tone, enterprise suitability, language correctness 5. Decision: PASS → status QUEUED (hand to M8) / FAIL → status QC_REJECTED (retain, optionally regenerate) 6. Persist Message (content, provenance, QC results, c...

M8 Rules (2)

M8-CONSTRAINTS-001
M8 sequence starts draw down M6-allocated budgets and respect M5 timing guidance. When global capacity is exhausted, starts queue by Decision Score (not FIFO). Closing-window urgency from M5 can reorder the queue.
Rationale: *#4 — Why does M8 enforce exit conditions as first-class logic? Because a sequence that ignores responses is spam and burns brand. Execution must halt immediately on reply, negative sentiment, window closure, or do-not-contact — exit logic is core to the module, not an afterthought.*

M8-EXIT-001
M8 enforces sequence exit conditions (reply, negative sentiment, window closed, do-not-contact, fatigue). Any exit immediately pauses/terminates remaining steps. Contact-level fatigue and Market_Interaction_History gates are checked before every send (CONTACT-GOVERNANCE-001).
Rationale: *#5 — Deepest reason M8 exists? Because execution is where strategy meets reality and brand capital is actually spent. M8 operationalizes timing and restraint — paced, condition-aware contact rather than a burst of messages. It is the discipline that the system always sends the right touch, to the right person, at the right cadence, or not at all.*

M9 Rules (4)

M9-CAPTURE-001
M9 captures recipient/third-party-observed actions as Engagement_Event entities (action_type, observed_via, occurred_at), distinct from M8's outbound Messages. Where traceable, each links triggering_message_id (ENGAGE-ATTRIBUTION-001).
Rationale: *#2 — Why treat engagement as a first-class signal that retriggers scoring? Because engagement IS inbound buying intent — a reply or repeated profile views should immediately update the account's score and may open or widen a window. Treating engagement as passive analytics would waste the strongest real-time intent signal the system has.*

M9-SIGNAL-001
High-intent Engagement_Events (reply, meeting-accept, repeated content interaction) retrigger M2 scoring and notify M5 to open/widen a Buying_Window. Engagement is a first-class signal class, processed near-real-time, not batched (ENGAGE-TRIGGER-001).
Rationale: *#3 — Why store engagement as discrete events rather than aggregate counts? Because timing and sequence carry meaning — three opens in an hour signal different intent than three over three weeks. Discrete, timestamped events preserve the temporal pattern (velocity, clustering, recency) that aggregates destroy.*

M9-GRANULAR-001
M9 stores discrete timestamped Engagement_Events, never pre-aggregated counters. Velocity/clustering/recency are computed at read time (ENGAGE-GRANULARITY-001). Sentiment on replies updates Contact.interaction_lineage and Market_Interaction_History.

M9-PRIMACY-001
Every recipient action that the system can observe is captured by M9. No other module infers engagement independently; M9 is the sole authority on engagement state, feeding all reactive and learning modules. ① Workflow Logic (step-by-step) 1. Engagement webhook/poll received from channel tools (open, click, reply, profile view, meeting accept) 2. Normalize → Engagement_Event; classify intent (HIGH/MEDIUM/LOW) 3. Link triggering_message_id where traceable; persist discrete event 4. If reply: run sentiment; update Contact.interaction_lineage + Market_Interaction_History 5. Emit to M8 (exit/advance), M2 (rescore), M5 (window), M11 (feedback) 6. If positive intent → trigger M10 human handoff path 7. If no engagement after N touches → emit learn...

ORCH Rules (3)

ORCH-NOLOGIC-001
n8n contains orchestration ONLY: routing, scheduling, integration. It must NOT contain scoring, reasoning, AI generation, conflict resolution, or any business logic — those live in services governed by the Rule_Registry (per the original n8n philosophy).
Rationale: *#3 — Why route everything through System_Events rather than direct calls? Because the event log is the system's audit spine and coordination mechanism at once. Routing via events means every cross-module action is traceable, causally linked, and reconstructable — and the denormalized caches can always be rebuilt from the log.*

ORCH-EVENT-001
All cross-module coordination flows through System_Events (immutable, causally linked per SYSEVENT-COORD-001). The orchestrator subscribes/publishes events; any cache must be reconstructable from the event log (SYSEVENT-PRIMACY-001).
Rationale: *#4 — Why does the orchestrator enforce the pipeline order (M0 first, etc.)? Because governance and reasoning gates only work if they cannot be bypassed. The orchestrator wires the mandatory sequence — M1 detect → M0 bless → M2 score → M3 enrich → M4 reason → M7 generate → M10 review (if gated) → M8 send → M9 engage → M11/M12 learn — so no stage can be skipped.*

ORCH-PIPELINE-001
The orchestrator enforces mandatory pipeline gates: no scoring before M0 blessing, no generation before M4 reasoning, no send before M8 (and M10 review where gated). These ordering constraints are wired in orchestration and cannot be bypassed by any module.
Rationale: *#5 — Deepest reason the orchestration layer exists? Because a system of independent intelligent modules needs a reliable, traceable nervous system to make them act as one coherent whole — without coupling them or hiding logic where it can't be governed. n8n is the connective tissue that keeps the system both modular and coordinated.*

Section 4: Integrations, APIs, Failure Handling (62 rules)

FAIL Rules (1)

FAIL-LEARN-01
Every failure generates a Failure Pattern Record: root cause, precursor signals, recovery effectiveness, and prevention strategy — failures are treated as lessons, not just events. ABM Business Logic Bible — Section 4 · Integrations, APIs & Failure Modes Rule ID Specification

GOV Rules (1)

GOV-ICS-001
CONFLICT RESOLUTION (FIN-C02): An account may not be promoted to HOT/TIER_1 unless its Intelligence Confidence Score (ICS) is >= 40. Below 40 the account remains in WATCH and enrichment continues. Human force-promotion below ICS 40 is permitted but triggers a logged SOFT WARNING (per AIHUMAN-DISAGREE-002). This is the rule previously referenced as GOV-001 in EXPLAIN-003. Rationale: the ICS>=40 floor was used consistently but its defining rule was unnamed.

INT Rules (37)

INT-ENR-01
Provider fallback chain is ordered Apollo → Clearbit → Clay; resolution stops at the first verified result. Order is configurable but the verified-only gate is not.

INT-ENR-03
Enrichment is gated by tier + reachability: HOT/WARM and path-reachable accounts enrich first; COLD/unreachable are deferred.

INT-ENR-04
Cross-provider conflict-merge: most-recent + highest-source-trust wins; unresolved title/role conflicts are flagged ~ and excluded from automated send.

INT-ENR-05
Every enriched record carries source + timestamp + freshness TTL; on expiry the contact is re-enriched before reuse.

INT-ENR-06
Buying-committee coverage is required: single-threaded enrichment (one contact per account) is flagged as insufficient coverage to scoring.

INT-ENR-07
Enrichment cost/volume is budgeted and logged; spend per account is capped and visible to the Scarcity (capital) layer. ABM Business Logic Bible — Section 4 · Integrations, APIs & Failure Modes

INT-LIN-02
Ban-risk circuit breaker: on early-warning signals (captcha challenge, view restriction, accept-rate collapse, withdrawal spike) the account is auto-paused and flagged for human review.

INT-LIN-04
Sending-account assignment matches the resolved SIG-PATH where a P1/P2 Decimal-person path exists; round-robin only for P4/P5 cold paths.

INT-LIN-05
Connection-accept and DM-reply update the contact state machine (→ OPENED/REPLIED) and trigger reply-detection + handoff logic.

INT-LIN-06
Account warmup: new sending accounts ramp action volume gradually before reaching standard caps.

INT-LIN-07
A banned/restricted account immediately emits SIG-PATH path-break events for every P1 edge it anchored. ABM Business Logic Bible — Section 4 · Integrations, APIs & Failure Modes

INT-EML-01
SPF, DKIM and DMARC must be configured and validated per sending domain before that domain is permitted to send.

INT-EML-02
Domain warmup is mandatory: new domains ramp volume on a schedule and cannot send at full volume until warmed.

INT-EML-03
Per-domain volume throttles + sending-identity rotation enforce a low-volume, high-relevance posture; mass blasting is structurally blocked.

INT-EML-04
Bounce, complaint and spam-placement telemetry feed a per-domain reputation score; below a floor, the domain is auto-paused from sending.

INT-EML-06
Reply received pauses the sequence and updates contact state (→ REPLIED) per FSD EXEC-002.

INT-EML-07
Duplicate-outreach prevention: the same contact is never sent the same touch twice across workflows (FSD EXEC-003).

INT-CRM-01
Sync is idempotent on (account_id, event_hash); re-delivery of an already-synced event is a no-op.

INT-CRM-03
Failed syncs route to a dead-letter queue with full payload + error; never a silent drop.

INT-CRM-04
On state conflict, CRM deal-stage wins and automated outreach to that account is paused pending human review.

INT-CRM-05
Reply / positive engagement triggers: pause automation → assign owner → generate AI meeting brief → Slack notify → create CRM task (FSD handoff sequence).

INT-CRM-06
CRM credentials are encrypted secrets; sync runs under least-privilege scopes (FSD security architecture). ABM Business Logic Bible — Section 4 · Integrations, APIs & Failure Modes

INT-SLK-01
Handoff notification fires in real time on the engagement trigger, carrying the AI meeting brief.

INT-SLK-02
Brief payload contract: account, triggering signal, resolved warm path/owner, and suggested angle — all grounded in INT-SIG facts.

INT-SLK-03
Notifications are deduplicated and priority-ranked; volume is rate-limited to prevent alert fatigue.

INT-SLK-04
Delivery is confirmed; an unconfirmed/failed notification is retried then escalated, never silently dropped. ABM Business Logic Bible — Section 4 · Integrations, APIs & Failure Modes

INT-ANL-01
Metabase reads from a read replica only; no analytical query touches the primary operational database.

INT-ANL-02
Analytics has no write access to operational tables; it is strictly read-only.

INT-ANL-03
KPI definitions are contracted and versioned (signal, persona, messaging, conversion performance) so the optimization loop tunes on stable metrics.

INT-ANL-04
Optimization outputs (weight/prompt changes) flow through a governed, reviewed, versioned loop — not applied directly from the BI layer. ABM Business Logic Bible — Section 4 · Integrations, APIs & Failure Modes

INT-ORC-01
n8n contains orchestration only: no business logic, no scoring logic, no AI reasoning. These live in versioned APIs/services.

INT-ORC-02
Every workflow step is idempotent; retries and re-runs cannot double-send outreach or double-write records.

INT-ORC-03
Workflows are grouped into Signal / Intelligence / Execution / Analytics domains with isolated failure boundaries.

INT-ORC-04
Every workflow follows the standard pattern: Trigger → Validation → Decision → Execution → Logging → State update.

INT-ORC-05
All external integration calls route through n8n so retry, rate-limit and observability policy are applied in one place.

INT-ORC-06
Workflow execution, decisions and errors are logged to the observability layer (OBS) on every run.

INT-ORC-07
n8n calls services over the defined API contracts (4.9); it never reaches into service databases directly. ABM Business Logic Bible — Section 4 · Integrations, APIs & Failure Modes

SIG Rules (23)

SIG-HYP-01

Every material signal stores competing explanations (hypotheses), each with a calibrated, RCM-stamped confidence; the engine reasons causally, not descriptively.

SIG-HYP-03

Hypotheses feed Strategy reasoning (Principle 5) as explicit causal premises, not hidden assumptions. ACCESS-01 ACCESS_STRENGTH is modelled separately from path existence (SIG-PATH), scored on relationship freshness, trust, reciprocity and introduction likelihood. ACCESS-02 Effective path rank = path tier × access strength; a P3 with strong access can outrank a P1 with weak/stale access. Sender assignment uses effective rank. ACCESS-03 Access strength is sourced from observable interaction signals only; it never asserts a private relationship

SIG-PATH-03

Every warm path carries a decay TTL. On expiry the path is re-resolved; an unresolvable previously-valid path emits a path-break event.

SIG-PATH-04

Executive departure from a target emits BOTH a target-vacancy intent signal (to SIG-EXEC/scoring) AND a follow-the-person event (re-anchor the relationship at the new employer).

SIG-PATH-05

Cross-group power-node detection: any individual resolved as holding roles across ≥2 target/priority entities is flagged as a high-value node and surfaced to every account they touch.

SIG-PATH-06

IT-subsidiary gatekeeper rule: where a target has a wholly-/majority-owned implementation arm (EJADA-equivalent), no C-suite outreach is sequenced until the gatekeeper relationship is established or explicitly waived by a human.

SIG-PATH-07

Mutual-connection confidence = f(shared-node count, shared-node seniority, recency of connection). Paths below a confidence floor are demoted to P4/P5.

SIG-PATH-08

Generalized connector registry (KPMG/Mastercard/Deloitte/Tarabut/Fintech-Saudi) is maintained as reusable P4 bridges with confirmed-reach metadata; a connector's reach is per-bank-verified, never assumed global.

SIG-PATH-09

No warm path above P5 → account is flagged 'reachable: cold only'; scoring applies an unreachability penalty so attention is not over-allocated to unreachable targets.

SIG-PATH-10

SIG-PATH never fabricates a path or a mutual connection. Unverifiable paths are recorded as ~inferred and excluded from automated sender assignment until human-confirmed. ABM Business Logic Bible — Section 4 · Checkpoint 4.1 · INT-SIG Account Intelligence Acquisition Engine 4.1.2 SIG-EXEC — C-SUITE DOSSIER & SPEECH CAPTURE SIG-EXEC — C-Suite Dossier & Speech Capture

SIG-EXEC-02

Public speeches, keynote quotes, interview statements are captured with source + date and stored as grounded personalization material linked to the named exec.

SIG-EXEC-03

Conflicting role-holders are flagged ■ verify and excluded from automated outreach until resolved; the engine never asserts one contested name confidently.

SIG-EXEC-04

Executive-move detection triggers: (a) dossier re-resolution, (b) target-vacancy intent signal, (c) SIG-PATH re-anchor / follow-the-person.

SIG-EXEC-05

Only SIG-EXEC-sourced, badged facts may be used as grounding for message personalization; ungrounded executive claims are rejected by QC. ABM Business Logic Bible — Section 4 · Checkpoint 4.1 · INT-SIG Account Intelligence Acquisition Engine 4.1.3 SIG-VENDOR & SIG-SUBS — VENDOR REGISTRY, GAP ANALYSIS & SUBSIDIARY ACTIVITY

SIG-VENDOR-01

Vendor registry is strictly per-account; a vendor fact confirmed at one bank is never propagated to another without independent confirmation (exclusivity discipline).

SIG-VENDOR-02
Each vendor entry carries a Decimal displacement/complement gap classification (displace · complement · no-fit) with source + freshness.

SIG-VENDOR-03
Vendor activity events (incumbent hiring surge, contract renewal/loss, case study) update the displacement-window state and may emit scoring triggers.

SIG-SUBS-01
Subsidiaries are resolved from the bank's official annual-report PDF (authoritative); subsidiary execs resolved to named persons (Phase-7 discipline) with badges.

SIG-SUBS-02
Subsidiary activity (launches, hires, regulatory moves) is ingested as account-linked signals; an IT/implementation subsidiary is additionally tagged as a SIG-PATH gatekeeper node. ABM Business Logic Bible — Section 4 · Checkpoint 4.1 · INT-SIG Account Intelligence Acquisition Engine 4.1.4 SIG-NEWS · SIG-EVENT · SIG-FIN · SIG-REG — DISCLOSURE & ACTIVITY STREAMS

SIG-NEWS-01
All four streams normalize + entity-resolve + deduplicate into one account intelligence graph; the same event from N sources collapses to one badged fact (badge rises with corroborating sources).

SIG-NEWS-02
Every ingested item carries source + date + a freshness TTL; expired items are demoted from active scoring but retained in Market Memory.

SIG-REG-01
Source-trust weighting: regulator (SAMA/CMA/Tadawul) > bank official > tier-1 press > aggregator > forum/rumour. Low-trust-only facts cannot cross the scoring-trigger threshold alone.

SIG-EVENT-01
Event/speaking detections cross-link to SIG-EXEC (capture the speech as grounded material) and to the Timing module (event proximity = engagement window). ABM Business Logic Bible — Section 4 · Checkpoint 4.1 · INT-SIG Account Intelligence Acquisition Engine 4.1.5 INT-SIG CONSOLIDATED RULE INDEX (34 new rules) All rules surfaced in this checkpoint, grouped by sub-stream. These are additive to the Section 1–3 rule base

Section 6: Knowledge-Acquisition Engine (19 rules)

ENR-COMP-01
Every potential enrichment action carries an Expected Information Gain estimate computed BEFORE execution. Finding a missing committee contact when the dossier has 2 known contacts has high estimated gain; finding the 8th has low. The estimate is the gate.

ENR-COMP-02
Enrichment STOPS when Expected Information Gain < Acquisition Cost — not when every dossier field is filled. Perfect dossiers are economically irrational. The 7-step workflow runs to completion only when each remaining step still clears the marginal-value gate.

ENR-BLOCK-01
Every dossier field is classified at design time as Blocking or Non-Blocking. Blocking: buying committee unknown, primary decision-maker unknown, no outreach path, no verified contact. Non-Blocking: secondary phone, office location, biography details.

ENR-BLOCK-02
Enrichment prioritization is Blocking-Information-First, not completeness percentage. The engine fills blocking gaps before any non-blocking, regardless of how easy a non-blocking field would be (closes the availability-bias trap).

ENR-BLOCK-03
Blocking-field gaps trigger LOW_COMMITTEE_COVERAGE and equivalent per-field flags; the flag binds §4.3 Reachability scoring and gates EXEC stage 6. Non-Blocking gaps do not gate downstream stages. ABM Business Logic Bible — §6 v3 (Information Economics) · Knowledge-Acquisition Engine

ENR-STOP-01
Every enrichment workflow carries an explicit Stopping Condition. Default for HOT: buying-committee coverage ≥ 80% AND ≥2 reachable contacts (Response Reachability per §6 v2) AND ≥1 outreach path resolved. When all three pass, enrichment stops.

ENR-STOP-02
Additional enrichment beyond the stopping threshold requires explicit justification: a named manual trigger or an event-driven trigger (§6 v2 ENR-REF-DEEP-01). Continuing past stopping by default is forbidden.

ENR-STOP-03
Stopping is observable: every enrichment cycle logs whether it stopped on the threshold (clean), on marginal-value gating (Layer 1), on cost ceiling (Layer 6), or on irreducibility (Layer 10). Stop-reason is a first-class telemetry field.

ENR-QUAL-03
A smaller, higher-quality dossier is preferred to a larger, lower-quality one. The 7-step workflow is configurable: a step that consistently produces low-Quality output for a segment may be skipped without completeness penalty. ABM Business Logic Bible — §6 v3 (Information Economics) · Knowledge-Acquisition Engine

ENR-DECAY-01
Every fact is tagged with one of four decay layers: Operational (7–30d), Tactical (30–90d), Strategic (6–12m), Structural (3–5y). Tag is set at acquisition; categories are not negotiable per fact.

ENR-DECAY-03
§6 v2 Iteration 3 Ramadan/Eid 1.5× decay-rate adjustment applies to Operational and Tactical, NOT Structural — organizational identity does not decay faster during Ramadan. ABM Business Logic Bible — §6 v3 (Information Economics) · Knowledge-Acquisition Engine

ENR-PORT-04
Vendor exclusivity discipline binds: per-account vendor facts (Backbase=SNB only, Efigence=Riyad only) are NEVER promoted into Shared Intelligence Assets. The Reusability classifier hard-rejects any fact tagged as bank-specific-vendor. ABM Business Logic Bible — §6 v3 (Information Economics) · Knowledge-Acquisition Engine

ENR-HUMAN-01
Research tasks (human and AI) must explicitly ask: what would disprove the current working hypothesis? The disconfirmation question is a required field in any structured research output for HOT accounts. Counters confirmation bias.

ENR-HUMAN-02
Research completion is measured by uncertainty reduction, not by field completion. A dossier with every field filled but no critical-unknown resolved is incomplete; a dossier with three fields filled that resolved all blocking unknowns is complete.

ENR-HUMAN-03
For major accounts (strategic or flagship tier), the dossier is not marked complete until a Contradictory Evidence Check has been performed and logged: 'we looked for evidence against the working narrative; here is what we found / didn't find.'

ENR-HUMAN-04
Availability bias counter: workflow order is set by Layer 7 (Dependency Value), not by what's easiest to find. The engine MAY override this order only with a logged justification.

ENR-BUDGET-01
Every identified unknown is classified at design time: Reducible (more research will likely resolve), Difficult (resolvable but expensive), or Currently Unknowable (no public source path; do not chase). Classification is a first-class field, not a comment.

ENR-BUDGET-02
Research stops on an unknown when Expected Cost of Further Reduction > Expected Value of the reduced uncertainty. Unknowable items stop immediately; Difficult items stop at their cost ceiling; Reducible items continue until the Layer 1 marginal-value gate binds.

ENR-REF-DEEP-01
). Continuing past stopping by default is forbidden.

Section 7: Trust and Influence Construction (27 rules)

AI Rules (24)

AI-NARR-01
Every account is assigned an Active Narrative (e.g. API Modernization, Digital Origination, AI Banking, Open Banking) derived from the dossier's Public Strategic Initiatives (§6 v2 ENR-INIT-DEEP-01). The narrative is the account-level story Decimal is telling, not a per-message choice.

AI-NARR-02
Agent 5 may vary the specific angle within a message but must remain inside the account's Active Narrative. A pivot to a different narrative requires an explicit account-level narrative change, not a per-message improvisation.

AI-NARR-03
Narrative drift — measured as the semantic distance between consecutive message angles for the same account — above a configured threshold triggers a review flag. Consistency is what creates trust; drift is observable and gated, not silently tolerated.

AI-SEQ-01
Agent 5 reads the contact's Interaction Stage: First Contact, Familiar, Engaged, or Active Discussion — derived from §10 engagement history (touch count, opens, replies) and contact_state (V2 §2.2).

AI-SEQ-04
Message optimization occurs at the sequence level: Agent 5's strategy brief states not just this message's angle but its role in the planned arc of touches, so a later message doesn't repeat an earlier message's role.

AI-TRUST-02
Aggressive CTAs (direct meeting asks, pricing discussions) are forbidden until the Trust Accumulation Score crosses a configured threshold OR the path carries high Trust Transfer Potential (AI-TRUST-04). Cold trust gets low-friction asks ('worth comparing notes?'); high trust gets direct asks ('15-minute call?').

AI-POL-01
Every personalization claim derived from an Inferred Buying Driver (§6 v2 ENR-INIT-DEEP-01) receives a Political Sensitivity Rating: Low, Medium, or High. The rating is computed strictly from SIG-TENSION's sourced friction signals (public statements, contradictory initiatives, budget conflicts) — NEVER from open-ended inference about internal politics. A claim with no SIG-TENSION basis defaults to Low sensitivity, not High; the engine does not invent risk.

AI-POL-02
High-sensitivity claims require human review regardless of the account's tier or the §8

AI-POL-03
When political risk is elevated (Medium or High), Agent 5 prefers Public Strategic Initiatives over Inferred Buying Drivers as the message's grounding — the safer, sourced, less politically-loaded angle is chosen, consistent with §6 v2 ENR-INIT-DEEP-03's anti-false-confidence default.

AI-POL-04
Political Sensitivity Rating is recomputed whenever the underlying SIG-TENSION signal changes; a rating is never assigned once and treated as permanent.

AI-REP-03
Agent 5's formal optimization objective is Expected Opportunity Value combined with Expected Reputation Value — not conversion probability alone. This is the engine's deepest objective-function change in §7 v2; every other layer in this document feeds into this combined objective.

AI-REP-05
Account-level success metrics (§13 Analytics) report BOTH Immediate Outcome (reply, meeting, opportunity) and Reputation Impact, never immediate outcome alone.

AI-REP-06
The engine may deprioritize a higher-conversion action in favor of a higher-reputation action ONLY when the Reputation Yield delta exceeds a configured floor (guardrail, not a vibe). Absent that explicit threshold being met, Agent 5 defaults to pursuing the highest-conversion action available — reputation is a constraint that can override conversion, but only when the trade-off is measurable and material, not as a default disposition. ABM Business Logic Bible — §7 v2 · Trust & Influence Construction System

AI-ATTN-01
Every drafted message receives an Attention Cost estimate — the mental effort required to process it, derived from length, structural complexity, and number of distinct ideas presented.

AI-ATTN-02
Agent 5 optimizes Attention-to-Value Ratio, not message length minimization alone. A 50-word high-relevance message with low cognitive effort beats a 300-word high-relevance message with high effort — this formalizes V2's existing 150/80-word caps as a ratio objective, not just a ceiling.

AI-ATTN-03
Different personas carry different Attention Budgets: C-suite (very low — shortest, most distilled framing); Director (moderate); Manager (higher — more detail tolerated). Agent 6's message complexity adapts to the target's seniority-derived attention budget, extending AI-AG6-07's seniority-matched framing rule with an explicit cognitive-cost dimension.

AI-INF-02
Stakeholder-specific objectives mean reducing legitimate friction through clarity and reassurance — never engineering around a stakeholder's legitimate concerns. 'Procurement Neutralization' means addressing real cost/process concerns with concrete, honest answers, not minimizing or routing around procurement's role. A message strategy that asks Agent 6 to obscure a real constraint is rejected by QC-007 (compliance check).

AI-INF-03
Messages to different stakeholders at the same account may pursue different objectives: CTO receives enthusiasm-building content; Compliance receives risk-reduction content (sourced, specific, never dismissive); Procurement receives friction-reduction content (clear process, pricing transparency). Same opportunity, different persuasion objective per role — this is the operational form of multi-threaded outreach already required by §4 (Reachability multi-thread bonus) and §6 (3–7 committee contacts).

AI-INF-05
A Stakeholder Influence Strategy assignment must be consistent with the Political Sensitivity Rating (Layer 4): a Procurement Neutralization message cannot rely on a High-sensitivity Inferred Buying Driver claim; it defaults to Public Strategic Initiative grounding per AI-POL-03. ABM Business Logic Bible — §7 v2 · Trust & Influence Construction System §7 v2 RULE ROLL-UP & PLACEMENT STATUS

AI-POL-001
binds strictly to SIG-TENSION's public, sourced friction signals — the engine never free-reasons about 'is this politically dangerous' from inference alone. Stakeholder-specific objectives are reassurance, not manipulation.

AI-AG6-07
's seniority-matched framing rule with an explicit cognitive-cost dimension.

AI-INF-002
defines 'Procurement Neutralization' and similar objectives as reducing legitimate friction through clarity, never as engineering around a stakeholder's legitimate concerns. The reputation-over-conversion trade-off is threshold-bound, not vibes-based.

AI-REP-006
requires a measurable Reputation Yield delta to exceed a configured floor before the engine may deprioritize conversion — otherwise Agent 5 defaults to pursuing the close.

AI-REP-003
makes this the formal optimization target for Agent 5. THREE GUARDRAILS MADE EXPLICIT IN THIS PASS Political sensitivity stays sourced-only.

ENR Rules (2)

ENR-INIT-DEEP-03
's anti-false-confidence default.

ENR-INIT-DEEP-01
). The narrative is the account-level story Decimal is telling, not a per-message choice.

QC Rules (1)

QC-HUMAN-004
post-MVP relaxation status. A High-sensitivity message never auto-sends, even after the 90-day relaxation criterion (§7 OPEN-Q7.1) is met for that segment.

Section 8: Trust Preservation and Irreversibility Management (19 rules)

QC-PHIL-01
Every QC rule (QC-RULE-001 through 010) maps to one or more Irreversible Risk Categories: Reputation, Relationship, Compliance, Market Memory, Political. A rule with no mapped category is incompletely specified.

QC-PHIL-02
QC severity = Probability × Magnitude × Reversibility Factor (0 for irreversible, approaching 1 for recoverable) — NOT raw frequency. A rare catastrophic failure (e.g. QC-RULE-009) gets highest protection despite low trigger rate.

QC-PHIL-03
Mapping: QC-001/002→Reputation; QC-004/005→Relationship; QC-006→Reputation; QC-007→Compliance; QC-RULE-008→Relationship+Market Memory; QC-RULE-009→Political+Compliance; QC-RULE-010→Compliance+Relationship.

QC-RISK-01
Each of the seven mechanical checks carries an explicit False Positive Cost (good message blocked, measured against §4 timing-window decay) and False Negative Cost (bad message sent, measured against Layer 1). Both are visible in QC configuration.

QC-RISK-02
QC optimization target for QC-RULE-001..007 is Expected Total Risk = (FP Rate × FP Cost) + (FN Rate × FN Cost), NOT maximum strictness. Thresholds may loosen where FP Cost is empirically shown to dominate.

QC-RISK-03
GUARDRAIL: applies ONLY to QC-RULE-001..007. Does NOT apply to QC-RULE-008/009/010, which remain strictness-maximizing regardless of false-negative cost. ABM Business Logic Bible — §8 v2 · Trust Preservation & Irreversibility Management

QC-HUMAN-08
Every QC reviewer carries a Reviewer Calibration Score: agreement rate with eventual outcomes, consistency across similar cases, and override-of-AI accuracy.

QC-HUMAN-09
Approval accuracy, override accuracy, and rejection accuracy tracked as THREE SEPARATE metrics per reviewer — a reviewer may excel at compliance issues but miss narrative-drift nuance.

QC-HUMAN-10
Periodic blind review audits: the same message routed to 2+ reviewers independently. Disagreement rate tracked; persistent disagreement on a check type triggers a calibration session, not blame.

QC-SYS-01
§8 tracks Failure Clusters: a statistically significant spike in any check type within a rolling window, segmented by account, segment, persona, or agent-version.

QC-SYS-02
A Failure Cluster breaching threshold triggers a System Health Investigation routed to engineering/prompt-ops, NOT per-message regeneration. Regenerating 200 failing messages treats symptoms; investigation finds the shared root cause.

QC-SYS-03
Failure-cluster telemetry is the engine's early-warning system: §8 sees every output from every upstream agent, so a rising hallucination rate is detectable here first. Binds to FAIL-LEARN-01 as a QC-specific instance of that broader pattern. ABM Business Logic Bible — §8 v2 · Trust Preservation & Irreversibility Management

QC-TRUST-01
Every QC failure is categorized by Trust Impact: Low, Medium, High, Critical — distinct from Irreversible Risk Category. Trust Impact measures damage to confidence IN THE SYSTEM, not the specific relationship.

QC-TRUST-02
§8 tracks a Trust Preservation Score per upstream module (Agent 5, Agent 6, §6 Enrichment, §4 Scoring, Narrative layer): rate of QC intervention weighted by Trust Impact. A declining score is the system's early signal that humans are about to start distrusting that module.

QC-TRUST-03
§8's success metric explicitly includes Trust Preserved, not only Messages Approved. §13 Analytics KPIs are extended to report Trust Preservation Score per module alongside QC pass rate — a high pass rate with declining trust score is a warning, not a success. ABM Business Logic Bible — §8 v2 · Trust Preservation & Irreversibility Management §8 v2 RULE ROLL-UP & PLACEMENT STATUS

QC-RULE-009
) gets highest protection despite low trigger rate.

QC-RULE-010
→Compliance+Relationship. LAYER 2 — FALSE POSITIVE / NEGATIVE BALANCE (RISK) Rule ID Specification

QC-RULE-008
→Relationship+Market Memory;

QC-RULE-001
through 010) maps to one or more Irreversible Risk Categories: Reputation, Relationship, Compliance, Market Memory, Political. A rule with no mapped category is incompletely specified.

Section 9: Dynamic Relationship Management (10 rules)

OUT-ATTN-03
Attention Availability inputs must be sourced from public signals only (SIG-REG deadlines, SIG-EVENT conference dates, SIG-NEWS earnings dates) — the engine never infers private calendar information about a named individual.

OUT-INTER-01
Every touch reads Sequence Memory — the full record of what this contact has already seen, across BOTH channels — before Agent 5/6 generate the next touch. This extends §7 v2 AI-NARR-01's Active Narrative to the per-contact touch history specifically.

OUT-INTER-02
Every touch receives a Reinforcement Score: does it strengthen the Active Narrative the contact has already been shown, or dilute it? A touch scoring as dilution is rejected and re-queued to Agent 5, using the same hard-gate mechanism as §8 QC-RULE-008 (narrative drift) — message interference is a specific, cross-channel instance of narrative drift.

OUT-RESP-03
Influence Signal feeds the Trust Accumulation Score (§7 v2 AI-TRUST-01) as an additional input category, NOT as a separate competing trust number — one trust ledger, richer inputs (Integration Decision 4).

OUT-RECIP-01
Every touch receives a Value Contribution Score: did this touch give the recipient something useful (an insight, a relevant benchmark, a specific observation) independent of any ask?

OUT-RECIP-02
Before a high-friction ask (direct meeting, pricing discussion), Cumulative Value Contribution across the sequence must exceed a configured threshold. This is the SAME gate as §7 v2

OUT-COMP-03
Major sourced competitor moves (SIG-VENDOR contract events, pricing news) become direct inputs to the active sequence's strategy brief, not merely account-dossier updates consumed at the next enrichment cycle — a live sequence can react within its current cadence.

OUT-COMPREL-03
The engine's optimization objective for sequence-level decisions is Lifetime Relationship Value, consistent with §7 v2 AI-REP-03's account-level Expected Reputation Value objective — extended down to the individual contact/sequence level so the two objectives are the same value function at different granularities, not two different goals. ABM Business Logic Bible — §9 v2 · Dynamic Relationship Deployment System

OUT-SEQ-08
, which already pauses on a Trust Accumulation drop — momentum makes the trend explicit rather than reactive only to an already-realized drop).

OUT-SEQ-03
(§9 v1) requires that a reply on ANY contact immediately pauses ALL sequences for the ENTIRE account.

Section 10: Reality Calibration Engine (8 rules)

ENG-EPIST-02
Observed Behavior and Inferred Intent are stored as separate fields on every engagement record, never collapsed into one. Downstream consumers (§4 scoring, §7 v2 Trust Accumulation) read Inferred Intent, never raw Observed Behavior directly — preventing the over-learning that comes from treating every click as equally meaningful.

ENG-NULL-03
Negative Signal Events are interpreted through Engagement Archetype (§10 v1 ENG-ARCH, §9 v2 OUT-BEH) exactly as ENG-SENT-06/ENG-SILENCE already require — this layer generates the event; archetype interpretation (already specified) consumes it. No new interpretation mechanism. ABM Business Logic Bible — §10 v2 · Reality Calibration & Institutional Learning Engine

ENG-TRUTH-01
Every significant engagement event is tagged with the specific prior belief(s) it bears on — named explicitly: a timing hypothesis (§5 v3 Patience layer), a persona hypothesis (§7 Agent 3), a narrative hypothesis (§7 v2 AI-NARR), or a trust hypothesis (§7 v2 AI-TRUST). An event with no identifiable prior belief it bears on is logged but does not feed any update.

ENG-TRUTH-02
§10's defined purpose is Model Correction, not Event Recording — this is the explicit architectural test for §10 rules going forward (mirrors §5 v3 TIER-ATTN-003 and §6 v3's information-economics test as the same kind of binding self-test). A proposed §10 rule that does not update some specific upstream belief is not a §10 rule.

ENG-LONG-02
Trend direction is read alongside current value in any downstream decision that uses Trust Accumulation or Engagement Score: a rising-40 and a falling-80 are NOT treated as equivalent inputs even when comparing two contacts at similar current values. This generalizes §9 v2

ENG-MEM-02
Patterns extracted from Learning Artifacts (e.g. 'this narrative angle underperforms with Compliance personas at Islamic banks') are stored as reusable organizational knowledge in Tier 6, NOT attributed to or siloed by the original account_owner — the knowledge survives personnel turnover by construction, because it lives in Tier 6, not in any individual's notes.

ENG-MEM-03
§10's defined output, per ENG-TRUTH-02's architectural test, includes its Tier 6 contribution as a first-class deliverable — a §10 implementation that logs events but never produces Learning Artifacts is incomplete, regardless of how complete its event-capture coverage is. ABM Business Logic Bible — §10 v2 · Reality Calibration & Institutional Learning Engine §10 v2 RULE ROLL-UP & PLACEMENT STATUS

ENG-SENT-06
/ENG-SILENCE already require — this layer generates the event; archetype interpretation (already specified) consumes it. No new interpretation mechanism.

Section 11: Human-AI Decision Partnership (14 rules)

HAND-COMP-01
Every handoff package leads with an Executive Summary Layer: a maximum 60-second read (~150 words) that sits ABOVE the full §11 v1 nine-component package, not instead of it. The full package remains available for the rep who wants depth; the summary is what most reps will actually read first.

HAND-COMP-02
Every insight in the package is ranked by Decision Importance (does this change what the rep should say or do in the next 10 minutes?), not by informational richness or recency. The richest, most recent enrichment fact does not automatically rank above an older, less detailed fact if the older fact has higher decision relevance — reuses §6 v3 Layer 4's Information Quality vs Quantity discipline (ENR-QUAL-01), applied to package ordering rather than enrichment acquisition.

HAND-COMP-03
The Executive Summary explicitly answers: 'if the rep remembers only three things, what are they?' — capped at exactly three items, forcing real prioritization rather than a longer 'top 5' that defeats the compression purpose.

HAND-JUDGE-01
Every suggested discussion angle (§11 v1 HAND-PKG component, sourced from §7 Agent 5) includes its Reasoning Chain in the package: not 'lead with regulatory urgency' alone, but 'lead with regulatory urgency because SIG-REG contributed the largest share of this account's §10 v2 ENG-CAUSE attribution for the triggering engagement.' The reasoning, not just the conclusion, transfers.

HAND-JUDGE-02
Every strategy recommendation in the package includes its Rejected Alternatives — the other angles Agent 5 considered and why each scored lower — so the human understands the tradeoff space, not just the winning choice. This is the package-facing view of the same comparison Agent 5 already performs internally; no new reasoning step, only added transparency. ABM Business Logic Bible — §11 v2 · Human-AI Decision Partnership Engine

HAND-DEC-01
Every §11 handoff that reaches a material next-action decision (which stakeholder to approach, which angle to use, whether to escalate) generates a Decision Record: options considered, selected option, rationale, assumptions, and — if the human's choice differs from the package recommendation — an explicit Override Reason selected from a defined taxonomy (local knowledge, relationship context, political consideration, timing concern, other-with-required-free-text).

HAND-DEC-04
Future reps and analysts can query Decision Records for a given account or persona type to see
Rationale: *prior decisions were made, not only what happened — institutional memory stores reasoning, retrievable independent of whether the original rep is still at the company. ABM Business Logic Bible — §11 v2 · Human-AI Decision Partnership Engine*

HAND-ACC-01
Every handoff records an explicit Decision Ownership Boundary with three separated stages: AI Recommendation (what §7 Agent 5 suggested and why), Human Decision (what the human actually decided, per the Layer 3 Decision Record), and Human Execution (how the decision was carried out in the actual conversation/email/call). Each stage is independently attributable.

HAND-ACC-02
Post-outcome reviews (§13 Analytics) classify failures into one of four categories: reasoning failure (the AI's underlying analysis was wrong — e.g. a hallucinated or miscalibrated fact), recommendation failure (the analysis was right but the strategic recommendation built on it was poor), execution failure (the recommendation was sound but human execution diverged or fell short), or external reality failure (everything upstream was reasonable; the outcome was driven by something genuinely unforeseeable — the §6 v3 ENR-BUDGET-01 'Currently Unknowable' category surfacing downstream).

HAND-BIAS-03
The package's design objective is explicitly to STIMULATE judgment, not replace it — this is an architectural test applied to every §11 v2 rule going forward, parallel to §5 v3's attention-allocation test and §10 v2's belief-correction test: a proposed §11 rule that makes the human's review faster but less critical fails this test. ABM Business Logic Bible — §11 v2 · Human-AI Decision Partnership Engine

HAND-KNOW-01
After any handoff interaction, the human can submit Missing Context Feedback (hidden sponsor identified, budget freeze mentioned, political concern surfaced, relationship detail) through a structured input — not a CRM free-text field that nothing downstream reads.

HAND-KNOW-02
Missing Context Feedback enters the engine through the EXISTING signal pipeline: it is created as a MANUAL-source signal (V2 §3.1's existing Manual Input source type) with the human as the named source, NOT a new ingestion mechanism. §3 SIG-DET normalization, dedup, and freshness rules apply to it exactly as they apply to any other signal.

HAND-CDL-03
§13 Analytics' success metric for handoff quality is explicitly Combined Decision Quality — the joint outcome of AI recommendation quality and human decision quality together — never AI quality or human quality scored in isolation. This is the §11-level instance of §8 v2's Trust Preservation principle: the system succeeds or fails as a partnership, not as two separately-graded components. ABM Business Logic Bible — §11 v2 · Human-AI Decision Partnership Engine

HAND-WF-07
already extended to account owners); an external reality failure updates nothing — it is logged and explicitly NOT used to penalize any component, consistent with

Section 12: Distributed Reality Coordination (21 rules)

CRM Rules (20)

CRM-TIME-01
Every sync conflict is classified as Temporal (both values were true at their respective timestamps; the disagreement resolves itself once the later event is applied) or Logical (the values are fundamentally contradictory regardless of timing — e.g. ABM=HOT and CRM=CLOSED_LOST cannot both be simply 'true at different times' in a way that self-resolves).

CRM-TIME-02
Temporal conflicts resolve automatically by applying the later Event Time's value (Layer 7) without escalation. Only Logical conflicts trigger INT-CRM-04's conflict-resolution escalation — this distinction is what prevents the unnecessary escalation volume that would otherwise drown the genuinely meaningful conflicts in noise.

CRM-REAL-01
Authority (per the Layer 6 Registry) means final decision rights on a field's VALUE, not superior knowledge about the underlying reality. A field's Authority Owner can be wrong; the Registry resolves WHO decides, not WHAT is objectively true.

CRM-REAL-02
Every conflict resolution explicitly logs the Lost Information — the losing side's value and its supporting context — even though it does not win. CRM wins deal-stage conflicts (INT-CRM-04), but ABM's discarded signal-based context is retained as a Tier 6 input, not deleted; it may still be useful even when it didn't carry the field. ABM Business Logic Bible — §12 v2 · Distributed Reality Coordination System

CRM-LAT-03
The engine distinguishes Unknown (no value has ever been recorded) from Outdated (a value exists but Freshness Confidence is low) as genuinely different states — an Outdated deal stage still gates automation per INT-CRM-04, but the handoff package (§11 v2 HAND-AUTH-02) surfaces 'last confirmed N days ago' rather than presenting it with the same confidence as a value updated an hour ago. ABM Business Logic Bible — §12 v2 · Distributed Reality Coordination System

CRM-FUSION-01
Every synchronized account maintains a Unified Reality Timeline: signals (§3), outreach touches (§9), engagement events (§10), handoffs and Decision Records (§11 v2), and CRM-pulled deal events (§12.2), merged chronologically by Event Time (Layer 7), not by which system happened to record them.

CRM-FUSION-02
§13 Analytics consumes the Unified Reality Timeline as its primary input for opportunity-history analysis, NOT isolated per-system event logs — a KPI computed from ABM events alone or CRM events alone is, by construction, working from a partial reality (Layer 2) and is flagged as such in any §13 report.

CRM-EPIST-01
The success metric for §12 is Reality Convergence, operationalized concretely as: a DECLINING Logical-conflict rate (Layer 1) on the same field over time, not merely 'sync completed' or 'fields match.' Two systems that agree because one blindly overwrote the other have NOT converged; two systems that increasingly agree because the underlying ambiguity was resolved HAVE.

CRM-OWN-01
Every synchronized field has exactly ONE declared Authority Owner: ABM, CRM, Governance, Human, or Derived (computed from other authoritative fields, itself carrying its own authority). No field has shared or co-equal ownership, ever.

CRM-OWN-02
A field with no declared Authority Owner CANNOT enter synchronization — this is enforced at the point a new field is added to the sync configuration, not discovered later when a conflict occurs.

CRM-OWN-03
The Authority Registry is a runtime-enforced artifact (not documentation): INT-CRM-02's existing ABM/CRM split and §11 v2 HAND-ACC-01's Decision Ownership Boundary (AI Recommendation / Human Decision / Human Execution) both register into it as existing entries, alongside every field added since. The Registry, not separate prose rules, is what §12's conflict-resolution logic actually queries at runtime. ABM Business Logic Bible — §12 v2 · Distributed Reality Coordination System

CRM-ORDER-01
Every synchronized event stores BOTH an Event Timestamp (when it actually happened, per the source system) and an Arrival Timestamp (when the ABM engine received it) as separate fields, never collapsed into one.

CRM-ORDER-02
State transitions are computed from Event Timestamp, never Arrival Timestamp. An event that arrives late but has an earlier Event Timestamp than the current state's last-applied event does NOT blindly overwrite — it triggers Layer 1's conflict classification against what's already been applied.

CRM-ORDER-03
A late-arriving event with an earlier Event Timestamp than already-applied events triggers State Reconciliation: the timeline (Layer 4) is recomputed with the correctly-ordered event inserted, and any state derived from the now-corrected sequence is re-evaluated — never a blind overwrite that could silently revert a correct later state to an incorrect earlier one. OPEN-DESIGN-QUESTION OPEN-Q12.3 — EVENT TIMESTAMP RELIABILITY ACROSS SOURCES Question: CRM-ORDER-01/02 assume every source can supply a trustworthy Event Timestamp. Not all sources are equal: HubSpot webhooks typically carry reliable event-time metadata; some signal sources (§3 INT-SIG) may only have a detection time, with the true underlying event time inferred or approximate. Wh...

CRM-AUDIT-01
Every field change stores Change Provenance: previous value, new value, actor (which system/person), timestamp, and reason where available — using the SAME append-only discipline as V2 §2.3 (signals) and SCORE-CALC-002 (scores): never overwritten, always a new historical record.

CRM-AUDIT-02
Every synchronized object supports Reality Replay: reconstructing the complete history of any field's values over time from its Change Provenance records, consumed by the Unified Reality Timeline (Layer 4) and available for compliance/audit requests.

CRM-AUDIT-03
No destructive updates anywhere in the CRM sync layer — this generalizes the append-only principle the Bible has used since §2 (SIGNAL, ACCOUNT_SCORE) to its full scope: every synchronized field, not only the two V2 originally called out. ABM Business Logic Bible — §12 v2 · Distributed Reality Coordination System

CRM-ECON-01
Every synchronized field has a Materiality Threshold below which a change does not trigger a push: e.g. a score change under 5 points does not push to HubSpot. The threshold is field-specific, not a single global cutoff (a tier CHANGE is always material; a score WOBBLE within a tier is not).

CRM-ECON-02
Sync is optimized for Decision Impact, not data freshness — reuses §6 v3 Layer 1's marginal-information-value discipline (ENR-COMP-02), applied to outbound propagation instead of inbound acquisition: push when the change would plausibly alter what a human does next, not whenever the underlying number ticks.

CRM-ECON-03
The materiality test is explicitly: 'will this change alter human behavior?' A tier change, a deal-stage-relevant state transition, or a new Decision Record clears this test readily; a sub-threshold score fluctuation does not. Sub-threshold changes still accumulate in Change Provenance (Layer 8) — they are not lost, only not pushed. ABM Business Logic Bible — §12 v2 · Distributed Reality Coordination System

SCORE Rules (1)

SCORE-CALC-002
) generalized to every CRM-synchronized field — not a new logging mechanism. Origin Metadata IS

Section 14: Opportunity State Machine (12 rules)

ENR Rules (1)

ENR-REACH-04
) AND an Agent-5-approved strategy brief exists AND QC is not in a blocked state.

STATE Rules (11)

STATE-GOV-01
§14 v1's STATE-ACCT-07 (EXEC Stage 5 sole writer of HOT-bound transitions) is GENERALIZED: every state transition in EITHER machine has exactly one declared Authoritative Writer. Examples: WATCHING→ENRICHING = EXEC Stage 5; HOT→OUTREACH_ACTIVE = §9 Outreach Orchestrator; ENGAGED→MEETING_BOOKED = Human + CRM (joint, per §11 v2 HAND-ACC-01's Decision Ownership Boundary); OPPORTUNITY→CUSTOMER = §12 v2's CRM Authority (INT-CRM-02). No transition exists without a named writer; this registers into the §12 v2 Authority Registry (CRM-OWN-03) as the state-transition instance of that general mechanism.

STATE-GOV-02
A state has MANY readers, exactly ONE writer — never multiple writers for the same transition. Where a transition appears jointly owned (e.g. ENGAGED→MEETING_BOOKED), it is specified as a single joint-writer pairing with an explicit protocol (human confirms, system records), not as two independent writers that could race.

STATE-GUARD-01
Every transition has explicit Entry Conditions beyond 'is this destination reachable from here.' Example: HOT→OUTREACH_ACTIVE requires enrichment complete (§6 ENR-STOP-01 stopping condition met) AND ≥1 Response-Reachable contact (§6 v2 ENR-REACH-04) AND an Agent-5-approved strategy brief exists AND QC is not in a blocked state.

STATE-GUARD-02
A transition request that fails its Entry Conditions creates a Transition Rejection Event (logged, not silently dropped) — retained for §13 v2 analytics (a recurring rejection pattern on the same transition is itself a Failure Cluster candidate, §8 v2 QC-SYS-01).

STATE-TIME-01
Every state in both machines has an Expected Residence Time: ENRICHING ≤ 48 hours; OUTREACH_ACTIVE ≤ 30 days (matches the 4-touch/14-day sequence plus buffer); HOT_UNENRICHED ≤ 14 days; (Opportunity states get their own residence times, specified in Part B).

STATE-TIME-02
Breaching Expected Residence Time triggers a State Health Alert — NOT an automatic transition. An account stuck in HOT_UNENRICHED for 20 days is flagged for human review, never auto-demoted or auto-retried indefinitely without visibility.

STATE-DER-02
Values reliably derivable from already-authoritative fields are NEVER additionally stored — this directly reduces §12 v2's sync-conflict surface (CRM-OWN-01's Authority Registry already requires exactly one owner per field; a derived field's 'owner' is simply the computation itself, with no write path to conflict).

STATE-OPP-06
The Opportunity State Machine (9 states above) exists INDEPENDENTLY of the Account State Machine (§14 v1, 13 states). They are never merged into one field, and no Opportunity state maps 1:1 onto an Account state.

STATE-OPP-07
Multiple Opportunities may exist simultaneously at one account, each independently positioned in the Opportunity State Machine — this is not an edge case to be handled, it is the NORMAL case the machine is designed for, consistent with §2's original re-anchoring rationale.

STATE-ACCT-07
(EXEC Stage 5 sole writer of HOT-bound transitions) is GENERALIZED: every state transition in EITHER machine has exactly one declared Authoritative Writer. Examples: WATCHING→ENRICHING = EXEC Stage 5; HOT→OUTREACH_ACTIVE = §9 Outreach Orchestrator; ENGAGED→MEETING_BOOKED = Human + CRM (joint, per §11 v2

STATE-OPP-02
) was honest about being temporary. Rule ID Phas Specification e

Section 15: Reliability, Resilience, Antifragility (15 rules)

FAIL-SEV-01
Every failure receives a Business Severity (P0 Critical / P1 High / P2 Medium / P3 Low), assigned INDEPENDENTLY of its §15 v1 failure type. The same failure type (e.g. CRM sync failure) can carry different severities depending on the specific object affected — a sync failure on a PROPOSAL-stage Opportunity is P0; the same failure type on a WATCHING account is P3.

FAIL-SEV-02
Retry urgency and recovery-queue priority are driven by Business Severity, not failure type alone — a P0 transient failure and a P3 transient failure both get exponential backoff per FAIL-02, but the P0 case escalates to human attention far sooner if backoff doesn't resolve it.

FAIL-CASC-01
Every failure stores its Upstream Dependency — what system or service it required to succeed. An enrichment-API outage that subsequently prevents contact discovery, which prevents outreach, which prevents engagement, which degrades scoring, is recorded as ONE root failure with a dependency chain, not four-plus unrelated incidents.

FAIL-CAP-02
The recovery queue is prioritized by Business Severity (Layer 1) × Opportunity Impact, NOT arrival order — the same ranking discipline as §5 v3 TIER-PORT-002 (Expected Opportunity Value ÷ Resource Cost), applied to recovery-item ordering rather than HOT-admission ordering.

FAIL-COST-01
Every P0/P1 failure receives a Business Impact estimate (opportunity delay, trust erosion, reputation risk, revenue risk) computed via the SAME Expected-Value family already used throughout the Bible (§6 v3 marginal-value estimation, §5 v3 TIER-PORT-002) — not a separate, third valuation method invented for failures specifically.

FAIL-COST-02
Operational reporting (§13 v2) includes Failure Cost (estimated Business Impact, aggregated) alongside Failure Volume — a high-volume, low-cost failure category and a low-volume, high-cost one are never collapsed into one 'failure rate' number.

FAIL-TRUST-01
Every failure is additionally classified by Reliability Impact (Low/Medium/High/Critical) — distinct from Business Severity (Layer 1), measuring damage to confidence IN THE PLATFORM specifically (will humans keep trusting the engine's outputs), mirroring §8 v2 QC-TRUST-01's Trust Impact taxonomy applied to infrastructure failures rather than QC failures.

FAIL-LAT-01
Every major failure tracks MTTD (Mean Time To Detect — failure occurrence to detection) and MTTR (Mean Time To Recover — detection to verified resolution, per Layer 7) as SEPARATE metrics, never blended into one 'incident duration' figure.

FAIL-LAT-02
Failure reviews prioritize closing detection bottlenecks BEFORE recovery bottlenecks when MTTD exceeds MTTR for a given failure category — a fast recovery process is wasted value if the failure isn't noticed for hours; FAIL-08's no-silent-failure principle (§15 v1) is necessary but not sufficient — 'logged' is not the same as 'detected promptly.'

FAIL-VERIFY-01
Every recovery action produces a Recovery Confidence Score — not merely a success/failure flag on the retry attempt itself. A CRM replay that executes without throwing an error has not necessarily verified that all records synced correctly, ordering was preserved (§12 v2 Layer 7), or no duplicates were introduced.

FAIL-VERIFY-02
Critical (P0/P1) recoveries require explicit Post-Recovery Validation — a check that the recovered state actually matches expected reality, not merely that the recovery action completed without throwing an exception.

FAIL-REPEAT-01
Every failure recurrence (matched against prior Failure Pattern Records, FAIL-LEARN-01) increments a Failure Recurrence Count for that specific failure signature, tracked across ALL §15 failure types — GENERALIZING §8 v2 QC-SYS-01 / §13 v2's Failure Cluster mechanism from QC-specific clusters to every category in §15. QC-SYS-01 is recognized as the earlier, narrower instance of this more general mechanism.

FAIL-REPEAT-02
Recurrence automatically escalates Business Severity (Layer 1): a 3rd occurrence of the same failure signature triggers a mandatory Root Cause Review regardless of its original severity rating — a P3 failure that recurs repeatedly is no longer treated as P3, because the recurrence itself is evidence of an unaddressed systemic issue.

FAIL-INVEST-02
Resilience fixes are prioritized by Expected Loss Prevented (reusing Layer 4's Business Impact estimation method), NOT by raw incident count — a rare P0 failure with high Business Impact can outrank a frequent P3 annoyance on the Register, exactly mirroring §5 v3 TIER-PORT-002's ratio-based ranking rather than a simple frequency sort.

FAIL-ANTI-02
Failure reviews explicitly answer: 'what new capability exists because this happened?' — a review that produces only a root-cause explanation, with no proposed capability improvement, is incomplete by this standard.

Section 16: Temporal Intelligence Layer (1 rules)

SCORE-SIG-01
) — a step-function signal's value should not be modeled as gradually fading when its real behavior is a cliff.

Section 17: Market Intelligence (KSA) (4 rules)

ENR Rules (2)

ENR-COMM-02
) remains the floor, but a thinner contact list with a strong connector graph is not penalized relative to a fuller list with no connector path.

ENR-COMM-01
) prioritizes resolving Sponsor status alongside the standard persona/seniority classification, not as an optional enrichment nice-to-have.

OUT Rules (1)

OUT-SEQ-EMAIL-02
) defaults Touch 1 to Awareness/Credibility for ALL accounts already;

SIG Rules (1)

SIG-POWER-01
) crosses a KSA-specific threshold.

Section 19: Glossary and Reference (1 rules)

QC-RULE-00
re-queues to STRATEGY (Agent 5), not wording (Agent 6) — because 8/009/010 the failure is in the plan, not the prose. Failure Cluster A statistically significant spike in one failure type, signaling an upstream §8 v2

WHAT DOCUMENT 2 (PRODUCTION) ADDS

Document 2 layers on top of this MVP foundation without contradicting a single rule here. It adds: full Opportunity-anchored dual-scoring, multi-channel outreach (Email + LinkedIn coordination), Autonomy Ladder A0-A2, EPIS Reality Confidence Model, the Resource Allocation Operating System (§5), Trust Construction and Preservation (§7-8), Dynamic Relationship Management (§9), Reality Calibration (§10), Decision Partnership (§11), Distributed Reality Coordination (§12), the Opportunity State Machine (§14), and Market Intelligence (§17). Total: 866 rules (462 MVP + 404 Production).